Integrated Communication Circuits

Lecture 6 : Frequency and Phase Modem

Prof. LIN MIN

ADVANCED SOC & IOT TECHNOLOGY LAB（ASITLAB）

第六章　角度的调制与解调

§6.1 调角波的性质

调频：FM ——— 高频振荡的频率受调制信号的控制

调相：PM ——— 高频振荡的相位受调制信号的控制

等幅波：$u(t)=U_m\cos\varphi(t)=U_m\cos[\omega(t)+\varphi_0]$

初始相位

$$\begin{cases} \omega(t)=\dfrac{d\varphi(t)}{dt} \\ \varphi(t)=\int\omega(t)\,\mathrm{d}t+\varphi_0 \end{cases}$$

一. 调频波、调相波：

1.调频波

$$\omega(t) = \omega_C + K_f u_\Omega(t)$$

瞬时角频率

未调制时载波的角频率

若设调制信号 $u_\Omega(t) = U_{\Omega m}\cos\Omega t$

$$\omega(t) = \omega_C + \boxed{K_f U_{\Omega m}}\cos\Omega t$$

$\Delta\omega_m$ → ω_C的偏移量

(最大角频率偏移与$U_{\Omega m}$成正比，与Ω无关)

$$\varphi(t) = \int_0^t \omega(t)dt + \varphi_0$$

$$= \omega_C t + \frac{K_f U_{\Omega m}}{\Omega}\sin\Omega t$$

令： $\varphi_0 = 0$

瞬时相位

m_f → 最大相位偏移，与Ω成反比

调频指数

表达式：$u(t) = U_m\cos(\omega_C t + m_f \sin\Omega t)$

6-1.exe

2. 调相波

$$u(t) = U_m\cos(\omega_C t + m_P\cos\Omega t)$$

$$\varphi(t) = \omega_C t + K_P u_\Omega(t)$$

设调制信号：$u_\Omega(t) = U_{\Omega m}\cos\Omega t$

6-2.exe

$$\varphi(t) = \omega_C t + \underbrace{\boxed{K_P U_{\Omega m}}}_{m_P}\cos\Omega t$$

最大相位偏移，与$U_{\Omega m}$成正比、与Ω无关

调相指数

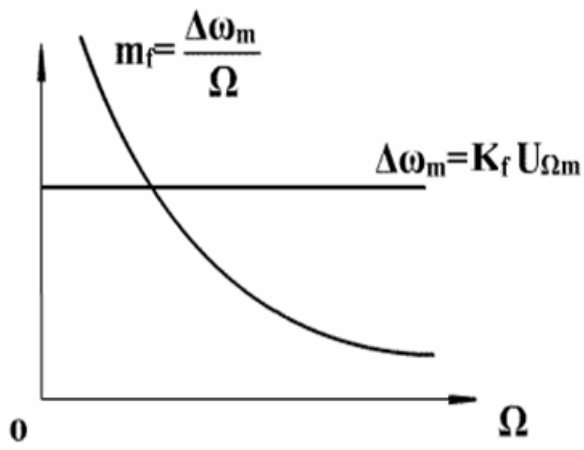
$$\omega(t) = \omega_C - \underbrace{\boxed{\Omega m_P}}_{\Delta\omega_m}\sin\Omega t$$

与Ω成正比

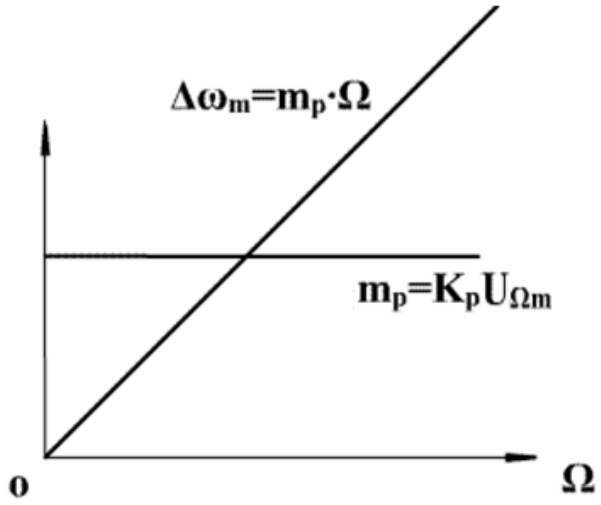
设调制信号 $u_\Omega(t) = U_{\Omega m}\cos\Omega t$ 情况下：

$$u(t) = U_m\cos(\omega_C t + m_f\sin\Omega t)$$ —— FM

$$u(t) = U_m\cos(\omega_C t + m_P\cos\Omega t)$$ —— PM



当 $U_{\Omega m}$ 恒定时，调频波



当 $U_{\Omega m}$ 恒定时，调相波

例1.

设有一组正弦调制信号，信号频率最低为 $F_{\min} = 300\text{Hz}$ 最高为 $F_{\max} = 3400\text{Hz}$。调制信号为等幅振荡。调频时，最大频偏 $\Delta f_m = 75\text{kHz}$；调相时，最大相偏 $m_p = 22\text{rad}$ 试求调频时调制指数 m_f 的变化范围及调相时最大频偏 Δf_m 的范围。

解

(1) 调频时： $U_{\Omega m}$不变，所以$\Delta\omega_m$不变，

而调频指数$m_f = \dfrac{\Delta\omega_m}{\Omega} = \dfrac{\Delta f_m}{F}$

$$m_{f\max} = \frac{\Delta f_m}{F_{\min}} = \frac{75\times10^3}{300} = 250$$

$$m_{f\min} = \frac{\Delta f_m}{F_{\max}} = \frac{75\times10^3}{3400} \approx 22$$

$\therefore$ 调频时，$\Delta\omega_m = k_f U_{\Omega m}$不变，

但$m_f \propto \dfrac{1}{\Omega}$,有很大变化。

(2) 调相时： $U_{\Omega m}$不变，所以m_p不变，

$$\Delta f_m = m_p F$$

$$\Delta f_{m\max} = m_p F_{\max} = 22 \times 3400 = 74.8\text{kHz}$$

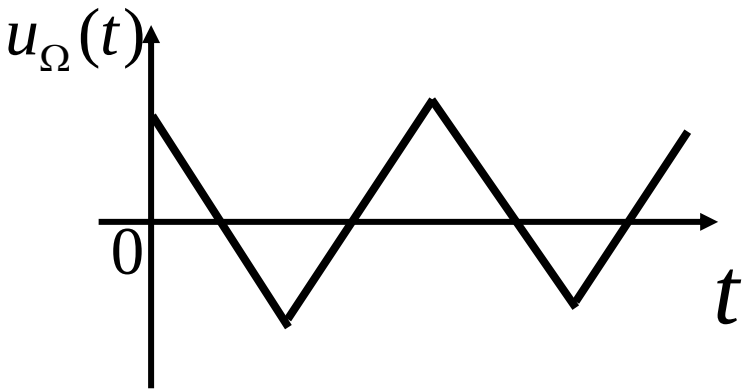
$$\Delta f_{m\min} = m_p F_{\min} = 22 \times 300 = 6.6\text{kHz}$$

$\therefore$ 调相时，$m_p = k_p U_{\Omega m}$不变

但$\Delta f_m \propto F$有很大变化

例2.

调制信号$u_\Omega(t)$ 为周期重复的三角波。画出调频波和调相波.



6-3.exe

6-4.exe

二. 调角信号的频谱

FM、PM波的数学表示式是相似的，统一写成调角波表示式

$$u(t) = U_m \cos(\omega_c t + m \sin \Omega t)$$

利用角函数展开：

$$u(t) = U_m [\cos(m \sin \Omega t) \cos \omega_c t - \sin(m \sin \Omega t) \sin \omega_c t]$$

在贝塞尔函数理论中，已经证明存在下列关系：

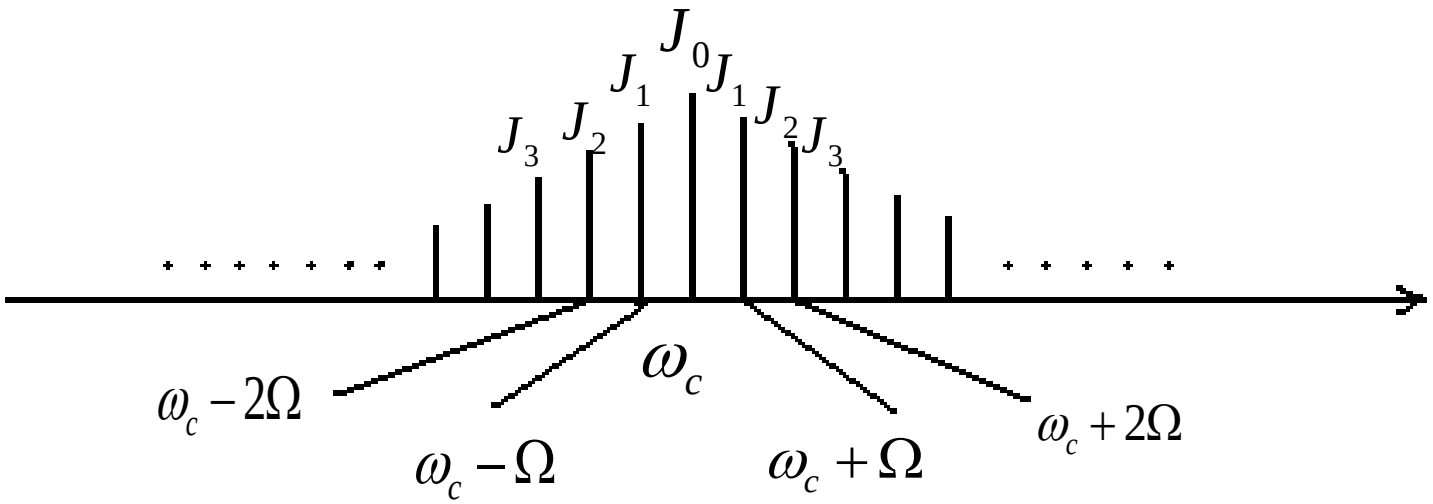
$$\cos(m \sin \Omega t) = J_0(m) + 2J_2(m) \cos 2\Omega t + 2J_4(m) \cos 4\Omega t + \ldots\ldots$$

$$\sin(m\sin\Omega t) = 2J_1(m)\sin\Omega t + 2J_3(m)\sin 3\Omega t$$
$$+ 2J_5(m)\sin 5\Omega t + \ldots\ldots$$

式中$J_n(m)$是以m为宗数的n阶第一 类贝塞尔函数。代入上式，整理得：

$$u(t) = U_m J_0(m)\cos\omega_c t + U_m J_1(m)[\cos(\omega_c + \Omega)t - \cos(\omega_c - \Omega)t]$$
$$+ U_m J_2(m)[\cos(\omega_c + 2\Omega)t + \cos(\omega_c - 2\Omega)t]$$
$$+ U_m J_3(m)[\cos(\omega_c + 3\Omega)t - \cos(\omega_c - 3\Omega)t] + \ldots\ldots$$

调角的频谱为：



1) $\omega_c \pm n\Omega$无限多对上、下边频分量 $U_m J_n(m)$

当n为偶数时，上、下边频分量相加

当n为奇数时，上、下边频分量相减

第一类贝塞尔函数

调制指数	载波	边频对												
m	J_0	J_1	J_2	J_3	J_4	J_5	J_6	J_7	J_8	J_9	J_{10}	J_{11}	J_{12}	J_{13}
0.00	1.00	-	-	-	-	-	-	-	-	-	-	-	-	-
0.25	0.98	0.12	-	-	-	-	-	-	-	-	-	-	-	-
0.5	0.94	0.24	0.03	-	-	-	-	-	-	-	-	-	-	-
1.0	0.77	0.44	0.11	0.02	-	-	-	-	-	-	-	-	-	-
1.5	0.51	0.56	0.23	0.06	0.01	-	-	-	-	-	-	-	-	-
2.0	0.22	0.58	0.35	0.13	0.03	-	-	-	-	-	-	-	-	-
2.4	0	0.52	0.43	0.20	0.06	0.02	-	-	-	-	-	-	-	-
2.5	-0.05	0.50	0.45	0.22	0.07	0.02	0.01	-	-	-	-	-	-	-
3.0	-0.26	0.34	0.9	0.31	0.13	0.04	0.01	-	-	-	-	-	-	-
4.0	-0.40	-0.07	0.36	0.43	0.28	0.13	0.05	0.02	-	-	-	-	-	-
5.0	-0.18	-0.33	0.5	0.36	0.39	0.26	0.13	0.05	0.02	-	-	-	-	-
5.52	0	-0.34	-0.12	0.26	0.40	0.32	0.19	0.09	0.03	0.01	-	-	-	-
6.0	0.15	-0.28	-0.24	0.11	0.36	0.36	0.25	0.13	0.06	0.02	-	-	-	-
7.0	0.30	0.00	-0.30	-0.17	0.16	0.35	0.34	0.23	0.13	0.06	0.02	-	-	-
8.0	0.17	0.23	-0.11	-0.29	-0.10	0.19	0.34	0.32	0.22	0.13	0.06	0.03	-	-
8.65	0	0.27	0.06	-0.24	-0.23	0.03	0.26	0.34	0.28	0.18	0.10	0.05	0.02	-
9.0	-0.09	0.25	0.14	-0.18	-0.27	-0.06	0.20	0.33	0.31	0.21	0.12	0.06	0.03	0.01
10.0	-0.25	0.05	0.25	0.06	-0.22	-0.23	-0.01	0.22	0.32	0.29	0.21	0.12	0.06	0.03

2) 当调制指数 m 增大时，则具有较大振幅的边频分量也增加。

3) 当m为某些特定值时，如m=2.405、5.520、8.653、……载波分量成分 $J_0(m)=0$。用来测量频偏 $\Delta\omega_m$ 和调制指数 m_f 。

三、调角信号的频谱宽度

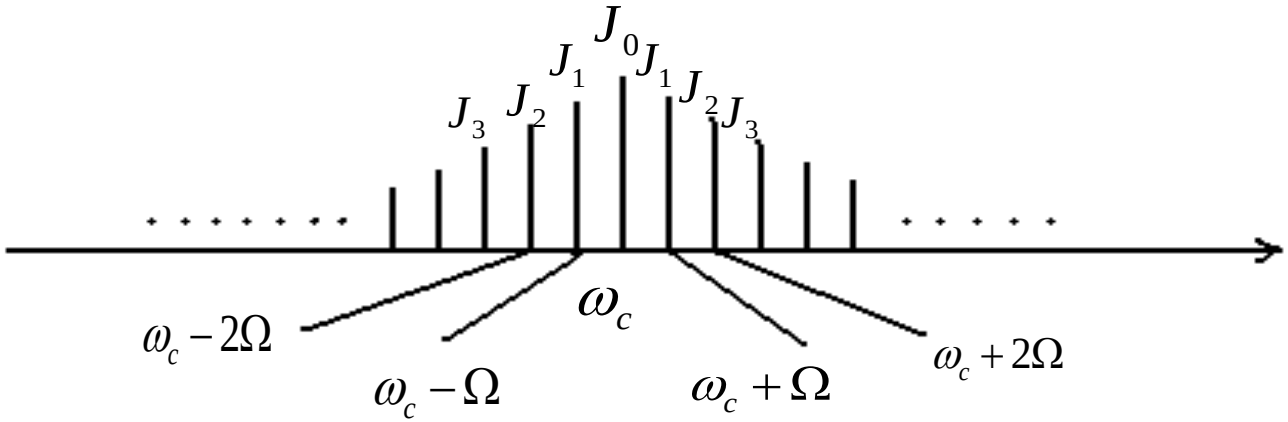
1) $m<1$,窄带调制 m很小，近似：$\cos(m\sin\Omega t)\approx 1$

$$\sin(m\sin\Omega t)\approx m\sin\Omega t$$

$$u(t)=U_m\cos(\omega_c t)+\frac{m}{2}U_m\cos[(\omega_c+\Omega)t]-\frac{m}{2}U_m\cos[(\omega_c-\Omega)t]$$

调角信号的频谱与AM波频谱相似。由ω_c、$\omega_c\pm\Omega$构成。差别就是下边频的相位相反。频谱宽度为 2Ω rad/s. 它广泛运用于移动通信中。

2) 宽带调制 $m>1$



频谱是无限宽的，认为当边频幅度小于载频的10%时，即忽略这些边频。因此，实际调角信号所占的有效频谱宽度是有限的。当$m>1$时，$m+1$ 以上各阶频谱的幅度均小于载波幅度10%。因而可以忽略。

有效频谱宽度为：

$$BW = 2(m+1)F = 2(\Delta f_m + F)\text{Hz} \qquad F = \frac{\Omega}{2\pi}$$

以上讨论的是单音调制情况。

实际上，调制信号中包括很多频率分量，设最高频率 $F_{\max}$

$$BW = 2(m+1)F_{\max} = 2(\Delta f_m + F_{\max})\text{Hz}$$

四. 各种调制方式的比较：

1. 抗干扰能力：

调角波的幅度不带有调制信号的信息，

抗干扰能力强.

2.信号频谱及带宽：

设调制信号的频率F： （m>1）

$$BW_{\mathrm{AM}} = 2F$$

$$BW_{\mathrm{FM、M、}} = 2(m+1)F$$

FM与PM的比较：

$$(BW)_{\text{FM、PM}} = 2(m+1)\Omega$$

但当Ω增加时，调频:

$m_f = \frac{K_f U_{\Omega m}}{\Omega}$减小，$BW_{\text{FM}} = 2(m_f+1)\Omega$略有增加；

调相:

$m_p = K_p U_{\Omega m}$不变，

$BW_{\text{PM}} = 2(m_p+1)\Omega$ 与Ω正比例增加.

3. 功率

AM: $$P_{平均} = (1 + \frac{1}{2} m_a{}^2) P_{载波}$$

$$m_a = 1: \quad P_{平均} = (1 + \frac{1}{2}) P_{载波} = 1.5 P_{载波}$$

FM、PM:

调角波振幅不变，工作在最大工作 状态.

功率: $$P_f = \frac{U_m{}^2}{2} \sum_{n=-\infty}^{\infty} J_n{}^2(m)$$

根据第一类贝塞尔函数特性：

$$\therefore P_f = \frac{U_m^{\ 2}}{2}$$

$$\sum_{n=-\infty}^{\infty} J_n^{\ 2}(m) = 1$$

上式表明，当Um一定时，不论 m_f 为何值，调频波的平均功率恒为定值，并且等于未调制的载波频率。换句话说，m_f 改变仅会引起载波分量和各边带分量之间功率的重新分配，但不会引起总功率的改变。

例3. 已知 $u_\Omega = U_{\Omega m}\cos 2\pi \times 10^3 t, m_f = m_p = 10,$

求(1)此时FM波和PM波的带宽?

(2)若$U_{\Omega m}$不变, F 增大一倍两种调制信号的带宽如何?

(3)若F 不变, $U_{\Omega m}$增大一倍两种调制信号的带宽如何?

(4)若$U_{\Omega m}$、F都增大一倍两种调制信号的带宽如何?

解:(1) $\because u_{\Omega} = U_{\Omega m}\cos 2\pi \times 10^3 t, \therefore F = 10^3\,\text{Hz}$

$$\text{FM}: BW_{\text{FM}} = 2(m_f + 1)F = 2(10+1)\times 10^3 = 22\text{kHz}$$

$$\text{PM}: BW_{\text{PM}} = 2(m_p + 1)F = 2(10+1)\times 10^3 = 22\text{kHz}$$

(2)FM: $U_{\Omega m}$不变，则$\Delta f_m = K_f U_{\Omega m}$不变；$m_f = \dfrac{\Delta f_m}{F} = 5$

$$BW_{\text{FM}} = 2(m_f + 1)F = 2(5+1)\times 2\times 10^3 = 24\text{kHz}$$

PM: $U_{\Omega m}$不变，m_p不变

$$BW_{\text{PM}} = 2(10+1)\times 2\times 10^3 = 44\text{kHz}$$

(3)FM：F不变，$U_{\Omega m}$增大一倍，

则$\Delta f_m = K_f U_{\Omega m}$增大一倍；$m_f = 20$增大一倍。

$$BW_{\mathrm{FM}} = 2(20+1)\times 10^3 = 42\mathrm{kHz}$$

PM：m_p也增大一倍,$(m_p = K_p U_{\Omega m} = 20)$

$$BW_{\mathrm{PM}} = 2(20+1)\times 10^3 = 42\mathrm{kHz}$$

(4)$F, U_{\Omega m}$均增大一倍，FM：m_f不变,$(m_f = \dfrac{\Delta f_m}{F})$

$$BW_{\mathrm{FM}} = 2(m_f+1)F = 2(10+1)\times 2\times 10^3 = 44\mathrm{kHz}$$

PM：m_p增大一倍，$m_p = 20$

$$BW_{\mathrm{PM}} = 2(20+1)\times 2\times 10^3 = 84\mathrm{kHz}$$

§6.2 调频的方法

直接调频： 对振荡器直接进行频率调制

（可获大频偏，但 f_C 不稳定）

间接调频： 对调制信号先积分后调相

（f_C 较稳定，但是频偏小）

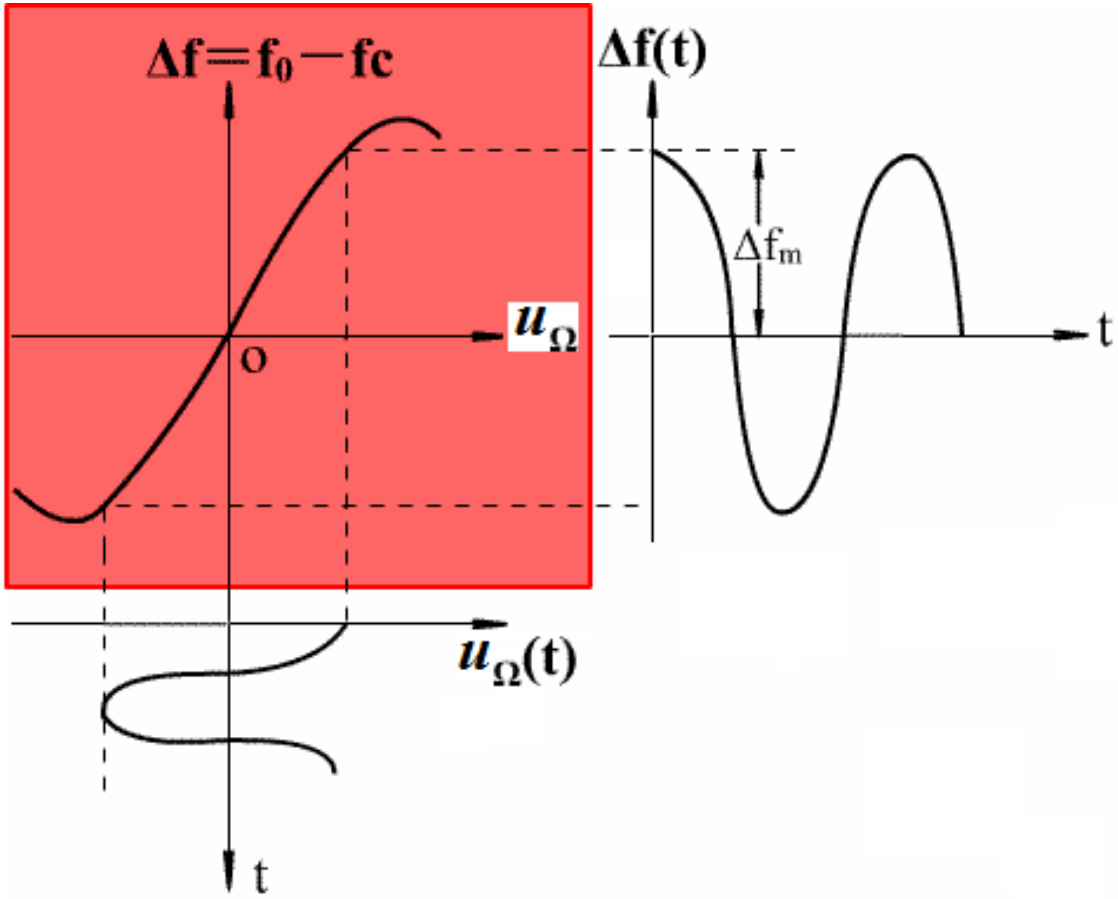
一. 调频电路的指标：

1. 调制特性

调频电路输出电压的频率偏移与调制电压的关系

（Δf——U_Ω）

要线性



2. 调制灵敏度：

单位调制电压变化所产生的频率偏移

要大 $$k_f = \frac{\Delta f}{u_\Omega}(\mathrm{Hz/V})$$

3. 中心频率（载波频率） **要稳定**

4. $\frac{\Delta f_m}{f_c}$ $> 1\%$ **大频偏**

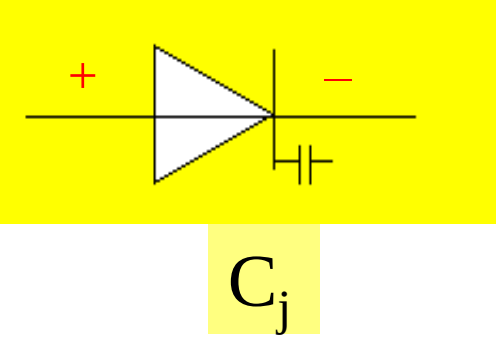
$< 1\%$ **小频偏**

二. 变容二极管直接调频电路：

利用它的结电容受反向外加电压控制而变化来实现调频。

1. 变容二极管

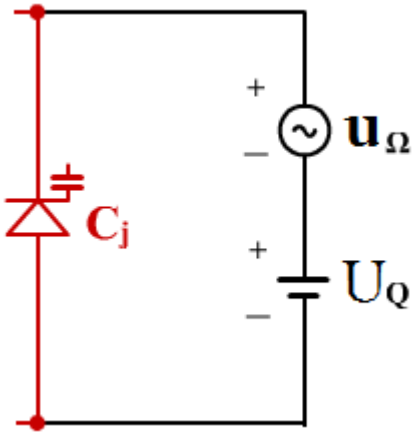
$$C_j = \frac{C_{j0}}{(1-\frac{u}{U_D})^{\gamma}}$$



C_{j0} —— 反向电压u = 0 时的结电容

U_D —— PN结内建电位差(很小)

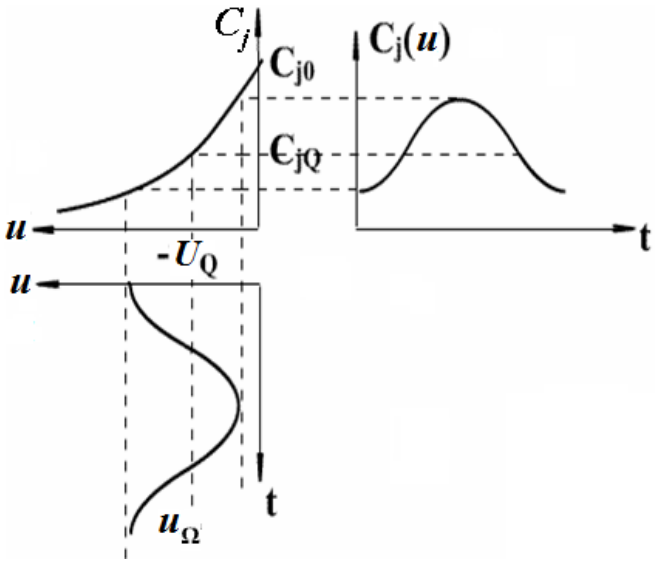
γ —— 电容的变化指数



当调制信号为：

$$u_\Omega(t) = U_{\Omega m}\cos\Omega t$$

$$C_j = \frac{C_{j0}}{\left(1+\dfrac{U_Q + U_{\Omega m}\cos\Omega t}{U_D}\right)^\gamma}$$



$$= \frac{C_{j0}}{\left(\dfrac{U_D + U_Q}{U_D}\right)^\gamma \left(1+\dfrac{U_{\Omega m}}{U_D + U_Q}\cos\Omega t\right)^\gamma}$$

$\rightarrow$ C_{jQ}

静态时的结电容：

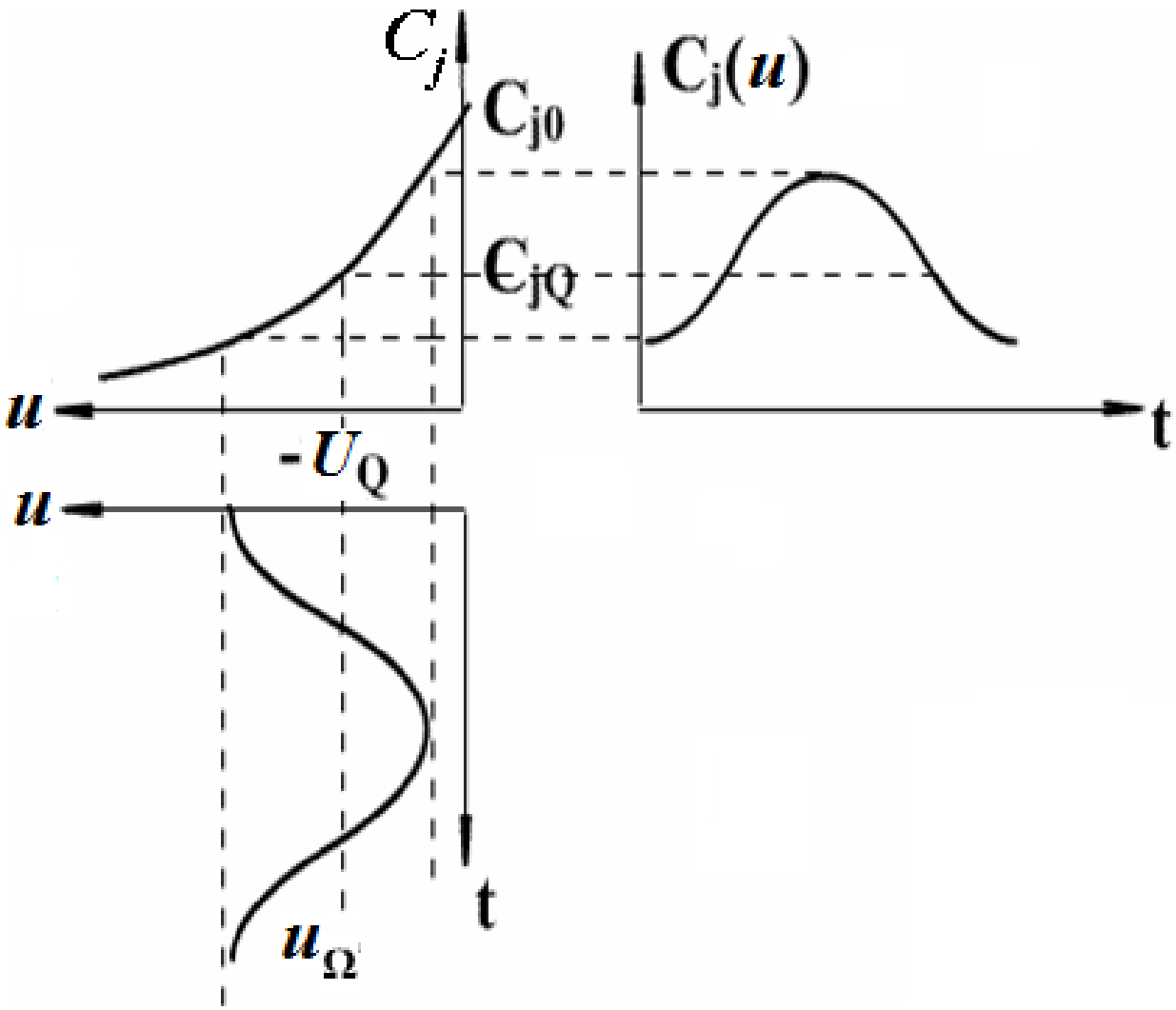
当$u=-U_Q$时，$C_{jQ}=\dfrac{C_{j0}}{(1+\dfrac{U_Q}{U_D})^{\gamma}}$

$$C_j=\frac{C_{j0}}{(\dfrac{U_D+U_Q}{U_D})^{\gamma}(1+\dfrac{U_{\Omega m}}{U_D+U_Q}\cos\Omega t)^{\gamma}}$$

$$C_j=\frac{C_{jQ}}{(1+M\cos\Omega t)^{\gamma}}$$

$$M=\frac{U_{\Omega m}}{U_D+|U_Q|}$$

结电容的调制度


C_j
C_{j0}
$C_j(u)$
C_{jQ}
u
$-U_Q$
t
u
t
u_Ω

2. 变容管作振荡回路的总电容

6-5.exe

大频偏直接调制电路

C_1 ——— 隔直作用

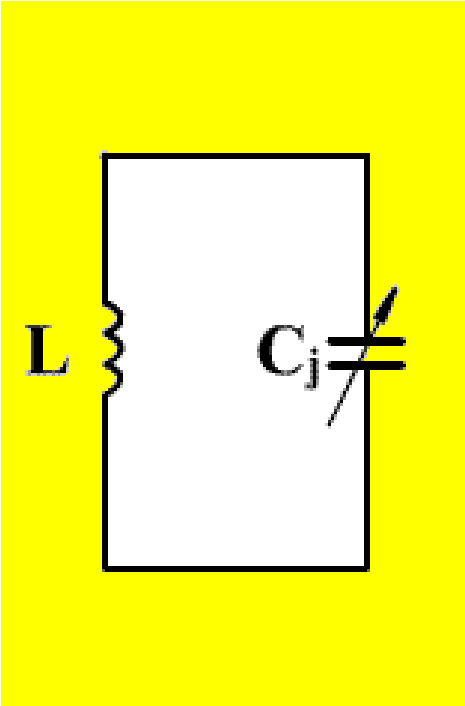
C_2 ——— 高频滤波电容

（对高频短路，对Ω开路）

L_1 ——— 高频扼流圈

应满足： $|U_Q| \geq U_{\Omega m}$

瞬时振荡角频率为:



$$\omega(t)=\frac{1}{\sqrt{LC_j}}=\frac{1}{\sqrt{L\dfrac{C_{jQ}}{(1+M\cos\Omega t)^{\gamma}}}}$$

$$=\omega_c(1+M\cos\Omega t)^{\frac{\gamma}{2}}$$

上式称为调制特性方程

$\omega_c=\dfrac{1}{\sqrt{LC_{jQ}}}$ 是静态工作点($u_\Omega=0$)的振荡角频率。

调制特性分析:

(1) $\gamma = 2$

$$\omega(t) = \omega_c + M\omega_c \cos\Omega t$$

线性调制

这时，调频波获得最大角频偏为 $\Delta\omega_m = M\omega_c$

(2) $\gamma \neq 2$

$$\omega(t) = \omega_c[1 + \frac{\gamma}{2} M \cos\Omega t + \frac{\gamma}{2}(\frac{\gamma}{2} - 1)\frac{1}{2!} M^2 \cos^2\Omega t]$$

$$= \omega_c[1 + \underbrace{\frac{1}{8}\gamma(\frac{\gamma}{2} - 1)M^2}_{\text{中心频率偏移}} + \underbrace{\frac{\gamma}{2} M \cos\Omega t}_{\text{基波分量}} + \underbrace{\frac{1}{8}\gamma(\frac{\gamma}{2} - 1)M^2 \cos 2\Omega t}_{\text{两次谐波分量}}$$

最大角频偏 $\Delta\omega_m = \frac{\gamma}{2} M\omega_c$

最大角频偏

$$\Delta\omega_{\mathrm{m}} = \frac{\gamma}{2} M\omega_{\mathrm{c}}$$

$$k_{\mathrm{f}} = \frac{\Delta f_{\mathrm{m}}}{U_{\Omega\mathrm{m}}} = \frac{\gamma}{2}\frac{Mf_{\mathrm{c}}}{U_{\Omega\mathrm{m}}}$$ 调制灵敏度

$$\Delta\omega_c \approx \frac{1}{8}\gamma(\frac{\gamma}{2}-1)M^2\omega_{\mathrm{c}}$$ 中心角频率偏离ω_{c}的数值

$$\frac{\Delta\omega_{\mathrm{c}}}{\omega_{\mathrm{c}}} \approx \frac{1}{8}\gamma(\frac{\gamma}{2}-1)M^2$$ 中心角频率的相对偏离值

当γ一定，增大M,可增大频偏$\Delta\omega_{\mathrm{m}}$,但同时也增大了$\Delta\omega_{\mathrm{c}}$和$\frac{\Delta\omega_{\mathrm{c}}}{\omega_{\mathrm{c}}}$

若M一定,提高ω_c,可增大调频波最大偏值$\Delta\omega_m$

例 调频振荡回路有电感L和变容二极管组成。$L=2\mu\text{H}$,

变容二极管的参数为: $C_{j0}=225\text{pF}, U_D=0.6\text{V}, \gamma=\dfrac{1}{2}$

$U_Q=-6\text{V}$, 调制$u_\Omega(t)=3\sin(10^4 t)\text{V}$, 输出FM波。求:

(1)载波f_c

(2)由调制信号引起的载波漂移Δf_c

(3)最大频偏Δf_m

(4)调频系数（调制灵敏度）k_f

(5)二阶失真系数

解

$$C_j = \frac{C_{j0}}{(1+\frac{u}{U_D})^{\gamma}} = \frac{C_{j0}}{(1+\frac{u_\Omega + |u_Q|}{U_D})^{\gamma}}$$

$$= \frac{225}{(1+\frac{3\cos 10^4 t + 6}{0.6})^{\frac{1}{2}}}$$

$$= \frac{C_{jQ}}{(1+M\cos\Omega t)^{\gamma}}$$

$$\because C_{jQ} = \frac{C_{j0}}{(1+\frac{U_Q}{U_D})^{\gamma}} = \frac{225}{(1+\frac{6}{0.6})^{\frac{1}{2}}} = 67.8\text{pf} \quad M = \frac{U_{\Omega m}}{U_D + U_Q} = 0.5$$

$$\therefore C_j = \frac{67.8}{(1+0.5\cos 10^4 t)^{\frac{1}{2}}}$$

$$\omega(t)=\omega_c(1+M\cos\Omega t)^{\frac{\gamma}{2}}\approx\omega_c+\Delta\omega_c+\Delta\omega_m\cos\Omega t$$

$$+\Delta\omega_{2m}\cos 2\Omega t$$

$$\omega_c=\frac{1}{\sqrt{LC_{jQ}}}=\frac{1}{\sqrt{2\times10^{-6}\times67.8\times10^{-12}}}\approx85.9\times10^6\,\text{rad/s}$$

$$\Delta\omega_m=\frac{\gamma}{2}M\omega_c=\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times85.9\times10^6\approx10.7\times10^6\,\text{rad/s}$$

$$\Delta\omega_{2m}=\Delta\omega_c=\frac{1}{8}\gamma(\frac{\gamma}{2}-1)M^2\omega_c$$

$$=\frac{1}{8}\times\frac{1}{2}\times(\frac{1}{4}-1)\times\frac{1}{4}\times85.9\times10^6\approx-10^6\,\text{rad/s}$$

$$(1)\ f_c=\frac{\omega_c}{2\pi}\approx13.7\text{MHz}$$

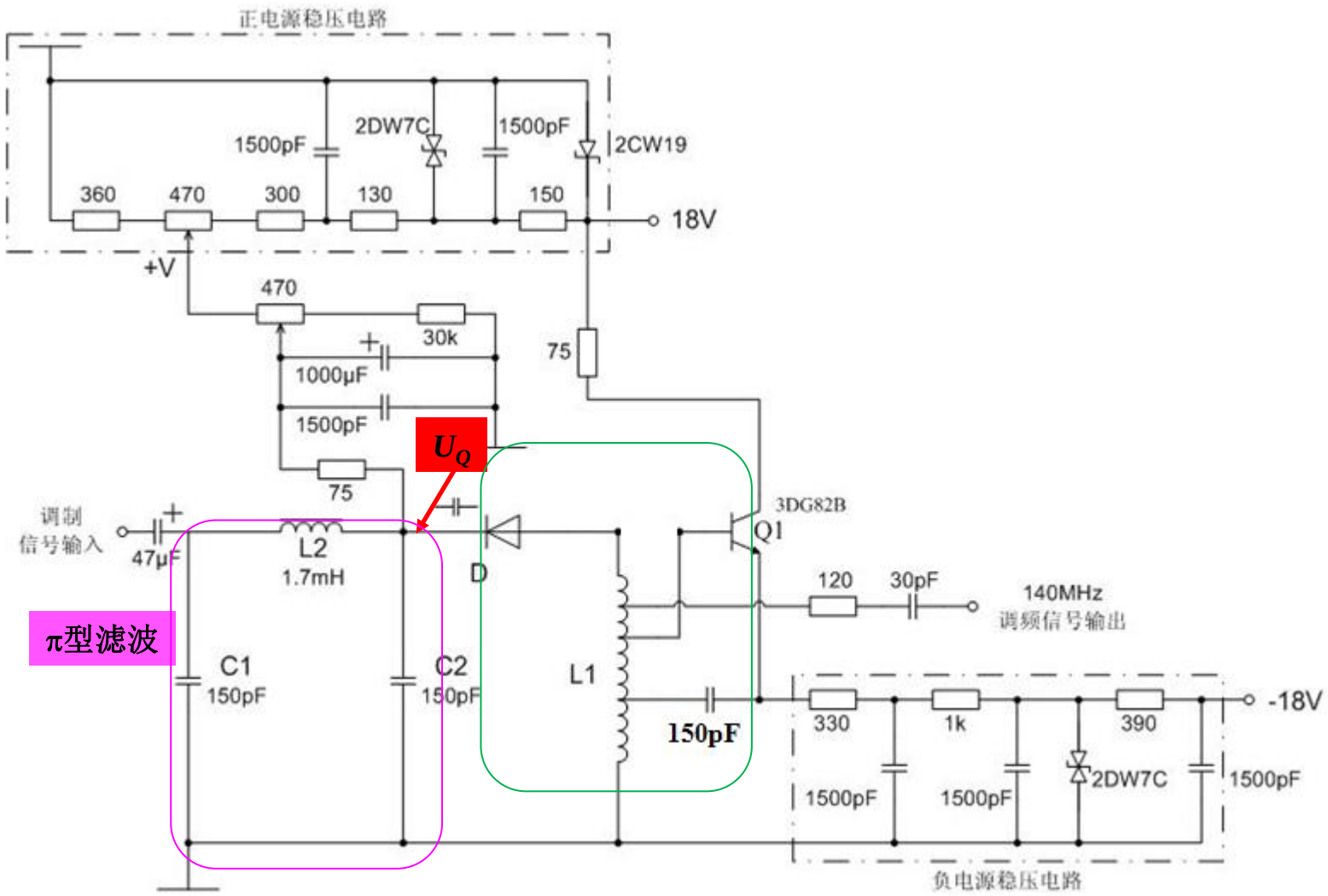
$$(2)\Delta f_c = \frac{\Delta\omega_c}{2\pi} = \frac{10^6}{2\pi} \approx 159\text{kHz}$$

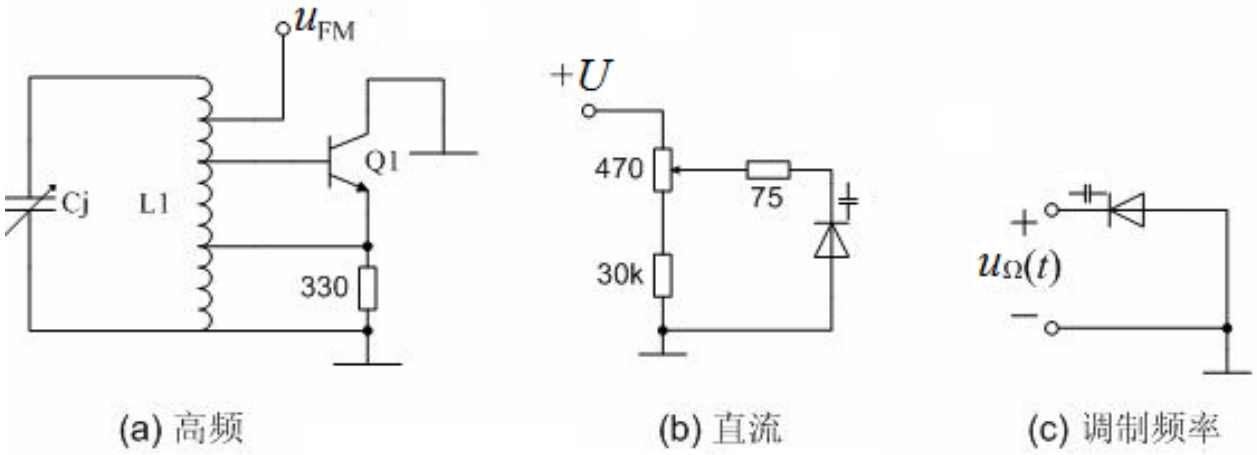
$$(3)\Delta f_m = \frac{\Delta\omega_m}{2\pi} = \frac{10.7\times10^6}{2\pi} \approx 1.7\text{MHz}$$

$$(4)k_f = \frac{\Delta f_m}{U_{\Omega m}} = \frac{1.7\times10^6}{3} = 5.7\times10^5\,\text{Hz/V}$$

$$(5)k_{f2} = \frac{\Delta\omega_{2m}}{\Delta\omega_m} = \left|\frac{M}{4}(\frac{\gamma}{2}-1)\right| = \left|\frac{1}{8}(\frac{1}{4}-1)\right| = 0.094$$

140MHz的变容管直接调频电路





- $L1$与变容管Cj构成振荡回路并与晶体管Q1接成电感三点式振荡电路。
- 变容管的直流电压偏置从正电源稳压电路中通过两个470Ω电位器取出一部分提供，作为U_Q。
- 调制信号通过1.7mH的高频扼流圈L_2和两个150pF电容C1、C2接成的π型滤波网络加到变容管上。

3. 变容管部分接入振荡回路

6-6.exe

小频偏的直接调频电路

C_3为高频滤波电容

$$C = C_1 + \frac{C_2 C_j}{C_2 + C_j} = C_1 + \frac{C_2 C_{jQ}}{C_2 (1 + M\cos\Omega t)^{\gamma} + C_{jQ}}$$

$$\omega(t) = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{L[C_1 + \frac{C_2 C_{jQ}}{C_2 (1 + M\cos\Omega t)^{\gamma} + C_{jQ}}}}$$

展开成幂级数，只考虑二次方项：

$$\omega(t)=\omega_c(1+A_1M\cos\Omega t+A_2M^2\cos^2\Omega t)$$

$$=\omega_c+\frac{A_2}{2}M^2\omega_c+A_1M\omega_c\cos\Omega t+\frac{A_2}{2}M^2\omega_c\cos 2\Omega t$$

$$\omega_c=\frac{1}{\sqrt{L(C_1+\frac{C_2C_{jQ}}{C_2+C_{jQ}})}}$$

$$A_1=\frac{\gamma}{2P}\qquad A_2=\frac{3\gamma^2}{8P}+\frac{1}{4}\frac{\gamma(\gamma-1)}{P}-\frac{1}{2}\frac{\gamma^2}{P(1+P_1)}$$

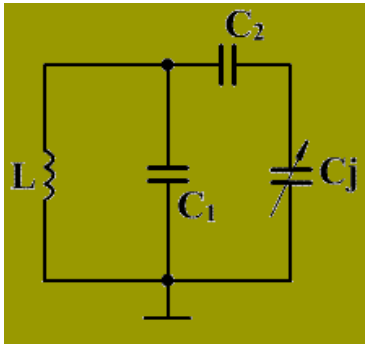
其中： $P=(1+P_1)(1+P_2+P_1P_2)$

$$p_1 = \frac{C_{jQ}}{C_2}$$

$$\frac{C_{jQ}}{C_{串}}$$

$$p_2 = \frac{C_1}{C_{jQ}}$$

$$\frac{C_{并}}{C_{jQ}}$$



最大频偏：$\Delta f_m = \frac{\gamma}{2P} M f_c$

中心频率偏移：$\Delta f_c = \frac{A_2}{2} M^2 f_c$

讨论：

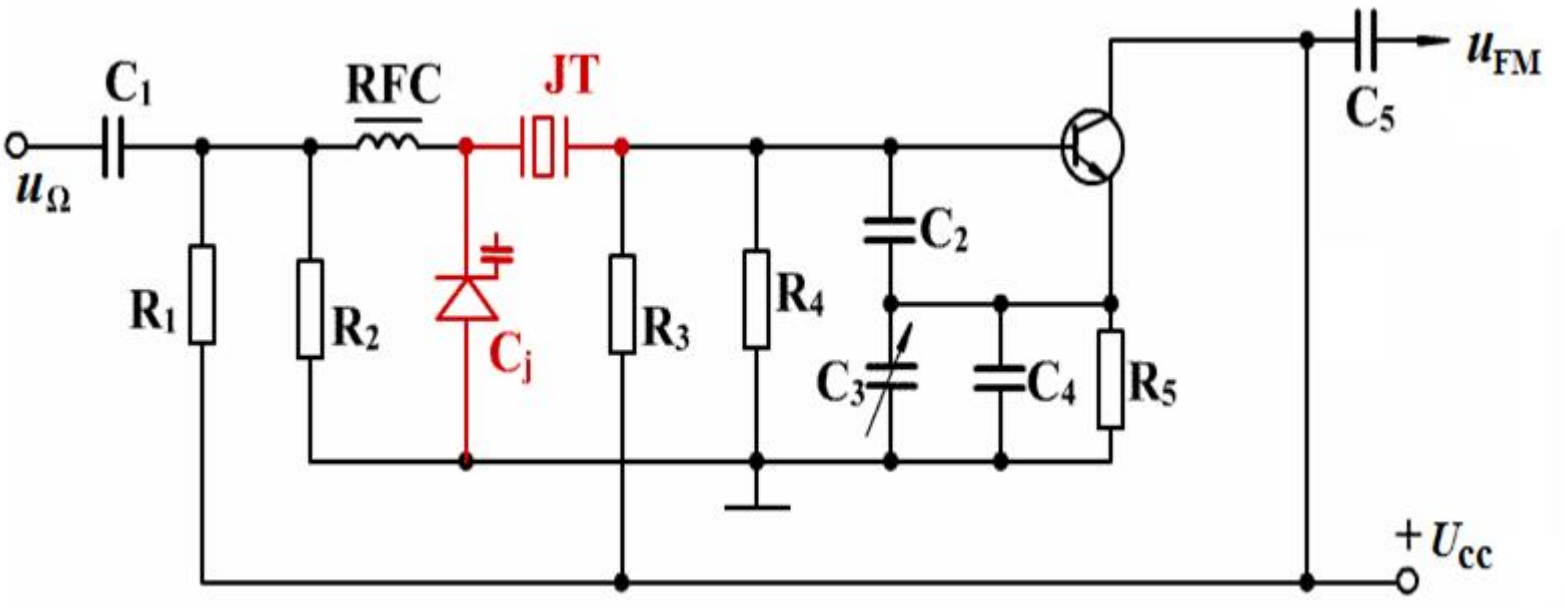
$$\left.\begin{array}{l} C_2\downarrow \quad P_1\uparrow \\ C_1\uparrow \quad P_2\uparrow \end{array}\right\} \longrightarrow P\uparrow \longrightarrow \Delta\omega_m\downarrow$$

C_1、C_2 的引入，使 C_j 对回路总电容的影响减小，从而 ω_c 的稳定性提高

例：题图所示电路采用2CC-401A变容管，已知变容管

$C_{j0}=109\text{pF}, U_D=1\text{V}, U_Q=-3.5\text{V}, \gamma=\frac{1}{2}$，石英晶体的$C_q=33.7\times10^{-3}\text{pF}$，音频调制到电压$U_{\Omega m}=1.4\text{V}$，振荡器的载频$f_c=12.5\text{MHz}$.

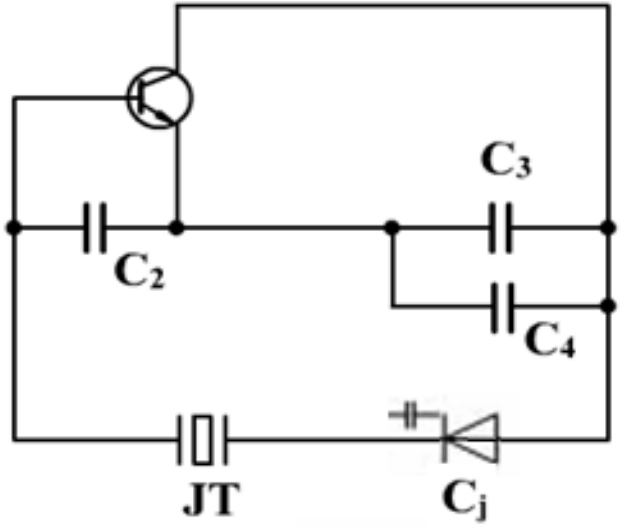
试指出各元器件的作用；画出高频等效电路；计算最大频偏Δf_m=?



各元器件的作用略。

解

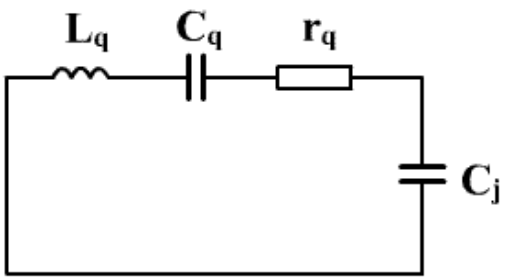
等效电路图



克拉泼振荡器

$$M=\frac{U_{\Omega m}}{U_D+|U_Q|}=\frac{1.4}{1+3.5}\approx 0.311$$

$$C_{jQ}=\frac{C_{j0}}{(1-\frac{U_Q}{U_D})^{\gamma}}=\frac{109}{(1+3.5)^{\frac{1}{2}}}\approx 51.4\text{pf}$$



$$P_2 = \frac{C_{并}}{C_{jQ}} = 0$$

由图可知：$P_1 = \dfrac{C_{jQ}}{C_{串}} = \dfrac{C_{jQ}}{C_q} = \dfrac{51.4}{33.7\times10^{-3}} \approx 1525$

则$P = 1 + P_1 = 1526$

最大频偏：

$$\Delta f_m = \frac{\gamma}{2P} M f_c = \frac{1/2}{2\times1526}\times0.311\times12.5\times10^6$$

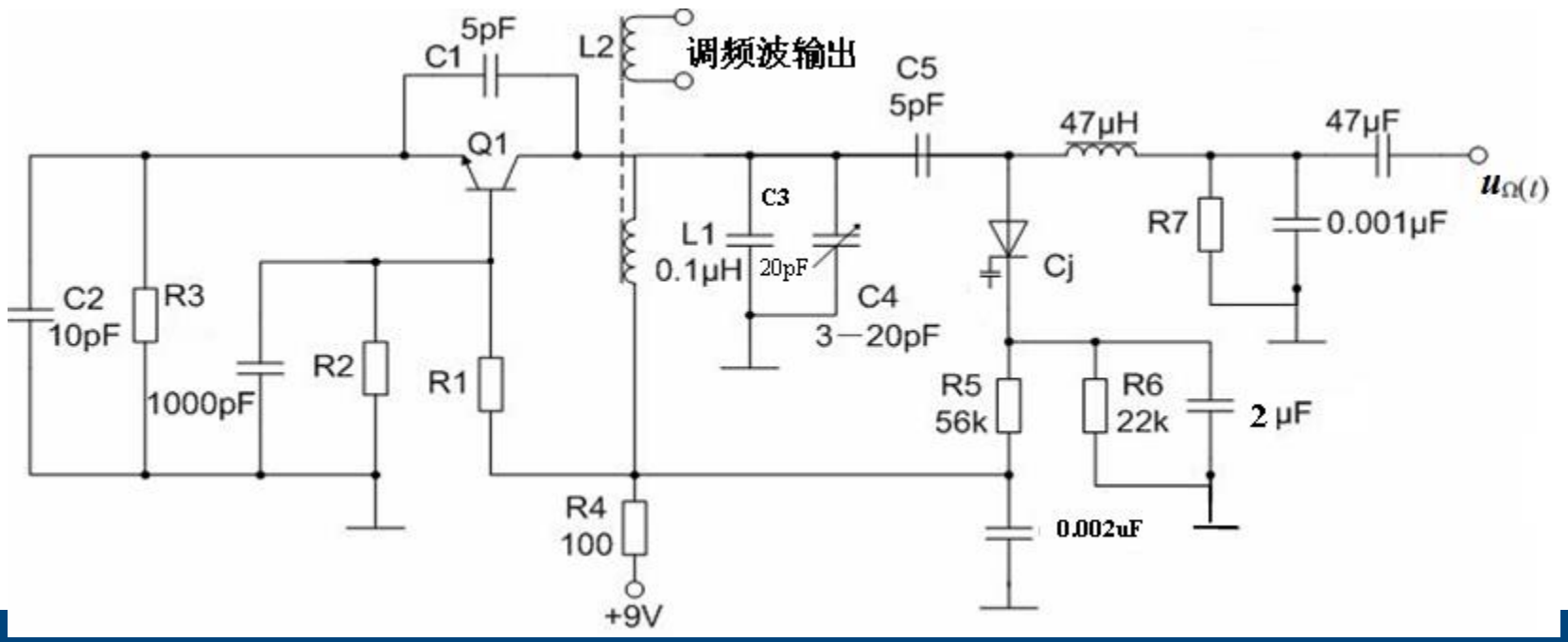
$$\approx 636.8\text{Hz}$$

例：对题图所示的变容管直接调频电路。

（1）说明电路中各元件的作用；

（2）画出该调频振荡器的高频通路、变容管的直流通路和音频通路。

（3）若已知 $u_\Omega = U_{\Omega m}\cos 2\pi\times 10^4 t(\mathrm{V})$，变容管结电容 $C_j = 80(U_Q + u_\Omega)^{-\frac{1}{2}}(\mathrm{pf})$，调频指数 $m_f = 5$，当 $u_\Omega = 0$ 时的振荡频率为 $f_c = 90\mathrm{MHz}$，试求变容管所需的直流偏置电压 U_Q，最大频偏 Δf_m 以及调制信号的振幅 $U_{\Omega m}$。



解

(1) R_1、R_2、R_3、R_4为晶体管提供直流偏置和直流通路;

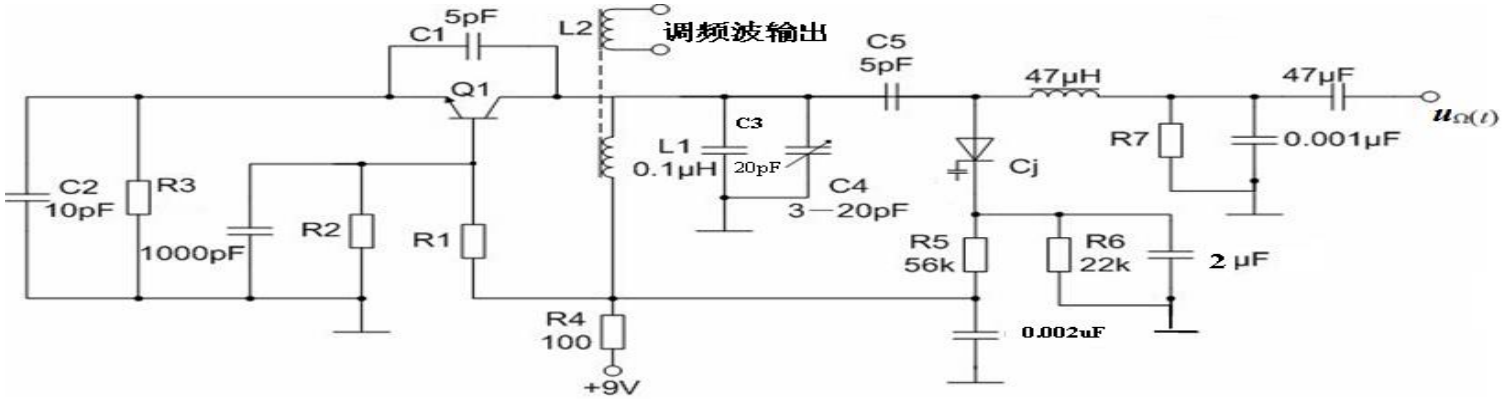
$R4$、R_5、R_6、R_7为变容管提供直流反偏置和直流通路;

1000pF，0.001μF、0.002μF均为高频旁路电容，即高频交流通路;

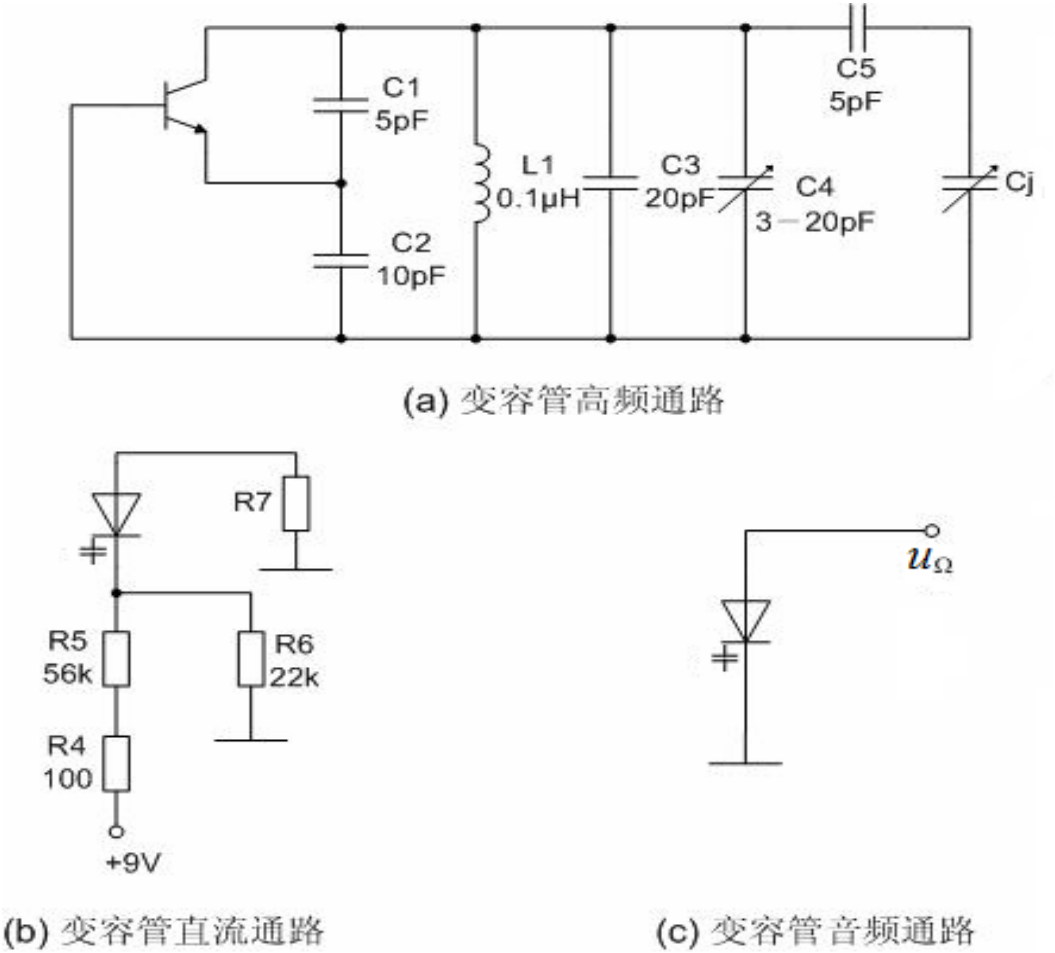
47μH为高频扼流图，对高频近似开路，对直流、低频近似短路;

47 μF、2 μF隔直电容，对低频近似短路;

C_1、C_2、C_3、C_4、C_5、C_j和L_1与晶体管Q1构成电容三点式振荡器； $L1$、$L2$互感耦合输出u_{FM}。



(2)



(a) 变容管高频通路

(b) 变容管直流通路

(c) 变容管音频通路

(3) $$U_Q \approx \frac{R_6}{R_5+R_6}\times 9\text{V} = \frac{22}{56+22}\times 9 = 2.54\text{V}$$

$$\therefore C_{jQ} = 80\Big/\sqrt{U_Q} = \frac{80}{\sqrt{2.54}} \approx 50\text{pF}$$

$$C_\Sigma = (C_1 串 C_2) 并 C_3 \ 并 C_4 \ 并 \ (C_5 串 C_j)$$

$$= \frac{C_1C_2}{C_1+C_2} + C_3 + C_4 + \frac{C_5C_j}{C_5+C_j}$$

$$\because \ f_c = \frac{1}{2\pi\sqrt{L_1C_\Sigma}} \qquad \therefore C_\Sigma = 31.3\times 10^{-12}F = 31.3\text{pF}$$

$$\therefore C_3 + C_4 = 23.5\text{pF}$$

$$P_1 = \frac{C_{\text{jQ}}}{C_5} = 10 \qquad P_2 = \frac{C_1 C_2 / (C_1 + C_2) + C_3 + C_4}{C_{\text{jQ}}} \approx 0.5$$

$$P = (1 + P_1)(1 + P_2 + P_1 P_2) = 71.5$$

由$\Delta f_{\text{m}} = \dfrac{\gamma}{2P} \text{M} f_{\text{c}}$ 而 $\Delta f_m = m_{\text{f}} F = 5 \times 10^4 \text{Hz} = 50\text{kHz}$

$$50 \times 10^3 = \frac{\frac{1}{2}}{2 \times 71.5} \times \text{M} \times 90 \times 10^6$$

得$\text{M} = 0.16$

又有$\mathrm{M}=\dfrac{U_{\Omega m}}{U_{\mathrm{Q}}+U_{\mathrm{D}}}\approx\dfrac{U_{\Omega m}}{U_{\mathrm{Q}}}$

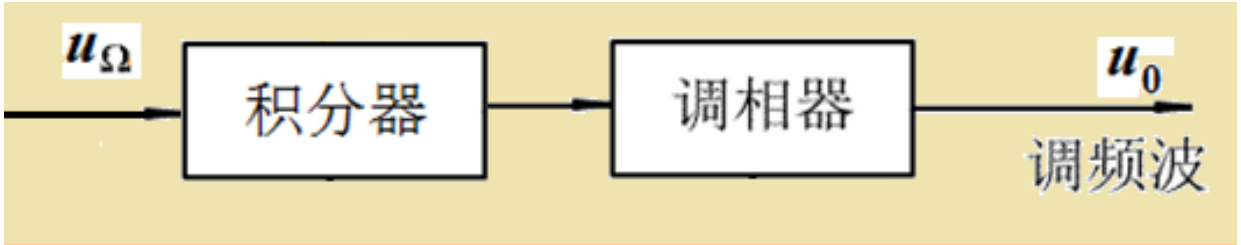
得$U_{\Omega\mathrm{m}}\approx MU_{\mathrm{Q}}=0.16\times2.54=0.41\mathrm{V}$

§6.3 间接调频电路

一. 间接调频的原理：用调相电路得到调频波

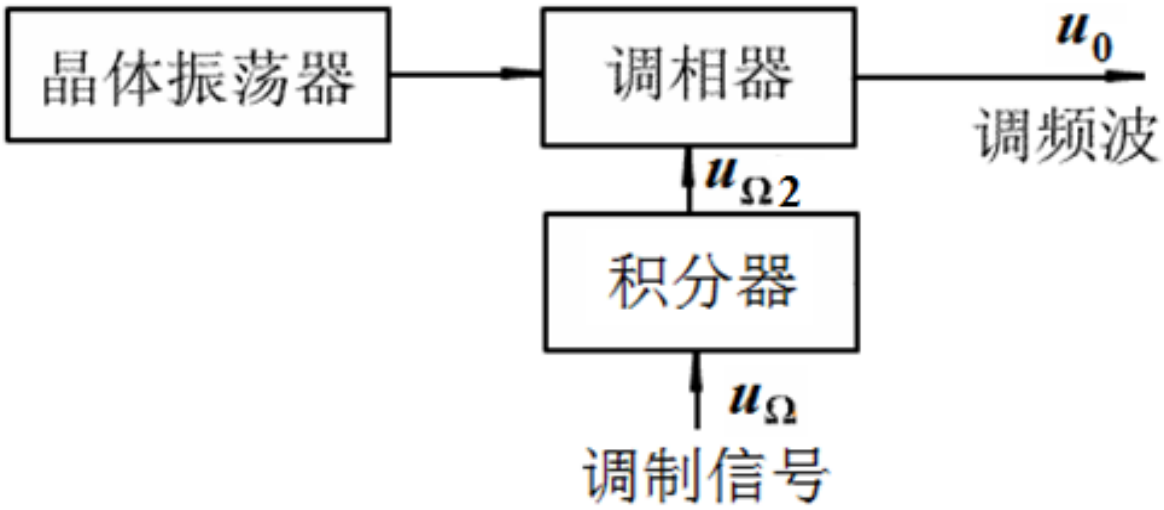
$$u_0(t) = U_m \cos(\omega_c t + k_p u_\Omega(t))$$

PM



$$u_0(t) = U_m \cos(\omega_c t + k_p \int u_\Omega(t) dt)$$

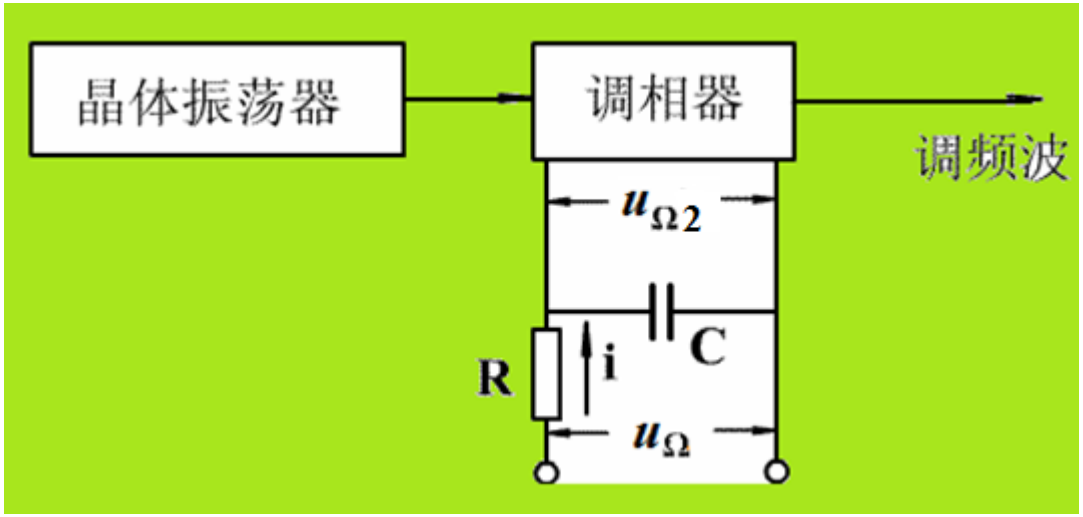
FM



$$u_0(t) = U_m \cos(\omega_c t + k_p u_{\Omega 2}(t))$$

$$u_0(t) = U_m \cos(\omega_c t + k_p \int u_\Omega(t) dt)$$

用 $u_{\Omega 2}$ 进行调相，产生的相位偏移为



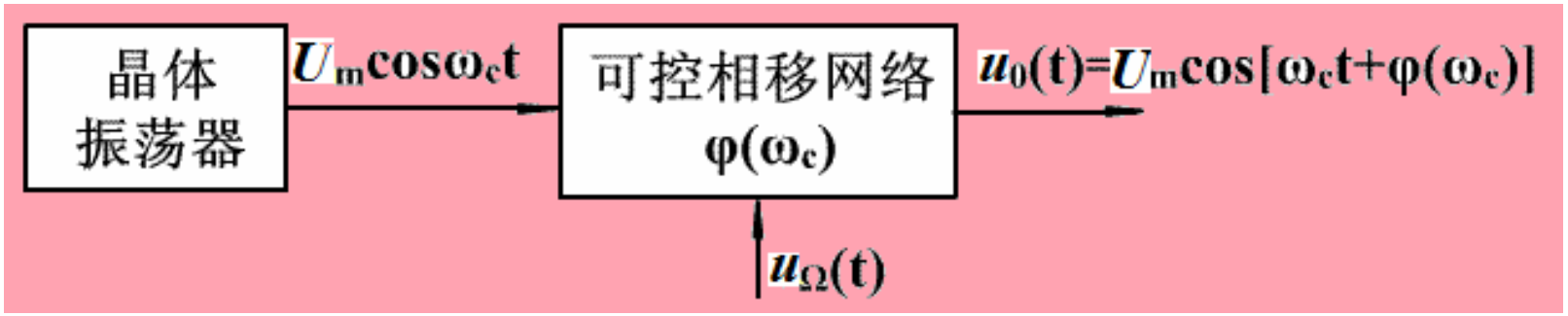
满足：ΩRC >> 1

设： $u_{\Omega} = U_{\Omega m}\cos\Omega t$ 则， $u_{\Omega} = Ri + \frac{1}{C}\int idt \approx Ri$

积分电路的输出电压， $u_{\Omega 2} = \frac{1}{C}\int idt = \frac{1}{RC}\int u_{\Omega}dt$

$$= \frac{1}{RC\Omega}U_{\Omega m}\sin\Omega t = U_{\Omega m2}\sin\Omega t$$

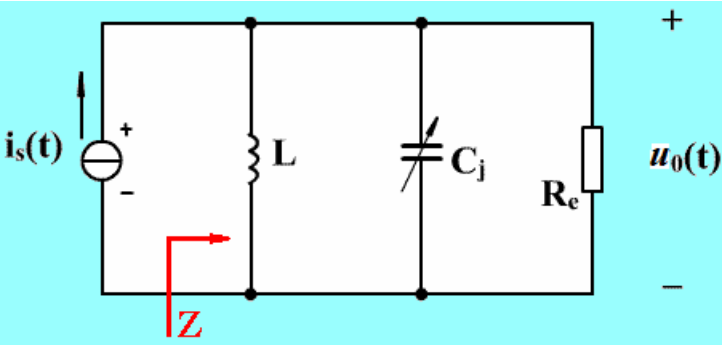
二. 调相电路



即, $\varphi(\omega_c)=k_p u_\Omega(t)=m_p\cos\Omega t$, 则相移网络的输出为调相波:

$$u_0(t)=U_m\cos[\omega_c t+\varphi(\omega_c)]$$
$$=U_m\cos(\omega_c t+m_p\cos\Omega t)$$

1. 变容管调相器



$$i_s(t) = I_{sm} \cos \omega_c t$$

$$Z(j\omega) = \frac{R_e}{1 + jQ_L \dfrac{2(\omega - \omega_0)}{\omega_0}} = |Z(\omega)| e^{j\varphi_z(\omega)}$$

$$\varphi_z(\omega) = -\mathrm{arctg}\left[Q_L \frac{2(\omega - \omega_0)}{\omega_0} \right]$$

其中： $\omega_0 = \dfrac{1}{\sqrt{LC_j}}$ $\quad Q_L = \dfrac{R_e}{\omega_0 L} \approx \dfrac{R_e}{\omega L}$

若加在变容二极管的电压： $u=-(U_Q+U_{\Omega m}\cos\Omega t)$

则有： $$C_j=\frac{C_{jQ}}{(1+\frac{U_\Omega}{U_D+U_Q})^\gamma}=\frac{C_{jQ}}{(1+M\cos\Omega t)^\gamma}$$

设： $u_\Omega=0, C_j=C_{jQ}, \omega_0=\omega_c=\frac{1}{\sqrt{LC_{jQ}}}$

$$\omega_0(t)=\omega_c(1+M\cos\Omega t)^{\gamma/2}\approx\omega_c(1+\frac{\gamma}{2}M\cos\Omega t)=\omega_c+\Delta\omega_0(t)$$

$$\Delta\omega_0(t)=\frac{\gamma}{2}M\omega_c\cos\Omega t$$

当$|\varphi_z(\omega)| < \frac{\pi}{6}$时，

$$\varphi_z(\omega_c) = -\mathrm{arctg}\left[Q_L \frac{2(\omega_c - \omega_0(t))}{\omega_0(t)}\right]$$

$$\approx -2Q_L \frac{\omega_c - [\omega_c + \Delta\omega_0(t)]}{\omega_c + \Delta\omega_0(t)} \approx \frac{2\Delta\omega_0(t)}{\omega_c} Q_L (\because \omega_c >> \Delta\omega_0(t))$$

$$= Q_L \gamma M \cos \Omega t = m_p \cos \Omega t$$

式中： $m_p = Q_L \gamma M, (m_p < \frac{\pi}{6})$

6-7.exe

因此，要实现不失真调相，除了选用γ= 2的变容管或限制M 为最小值，还必须限制$m_p < \frac{\pi}{6}$rad。

根据上述原理构成的电路:

6-8.exe

当$u_\Omega(t)=U_{\Omega m}\cos\Omega t$时，

$$u_\Omega'(t)=\frac{1}{C_4}\int_0^t i_\Omega(t)dt=\frac{1}{R_3C_4}\int_0^t u_\Omega(t)dt=\frac{U_{\Omega m}}{\Omega R_3C_4}\sin\Omega t$$

其中，$M=\dfrac{U_{\Omega m}}{\Omega R_3C_4(U_Q+U_D)}=\dfrac{U_{\Omega m}'}{U_Q+U_D}$

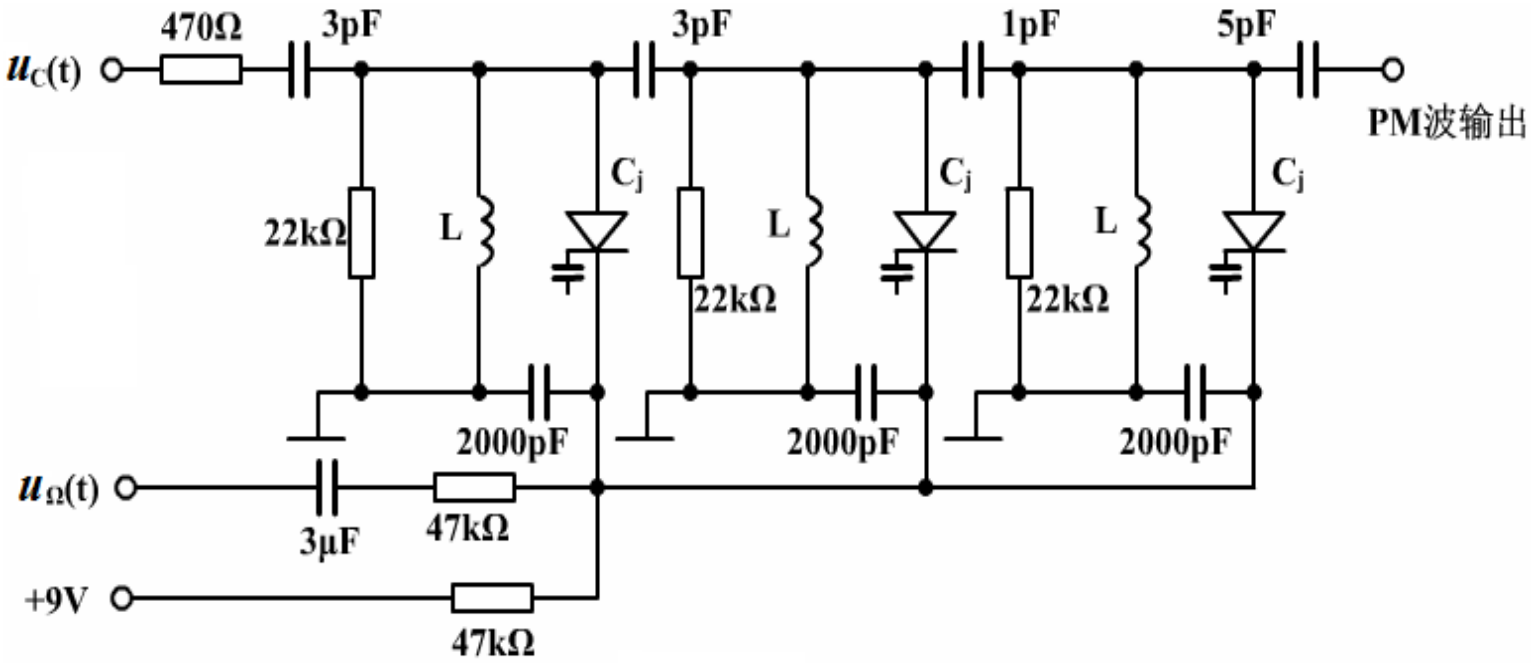
调相电路便转为间接调频电路。

且，$m_f=m_p=\dfrac{U_{\Omega m}Q_L\gamma}{\Omega R_3C_4(U_Q+U_D)}$

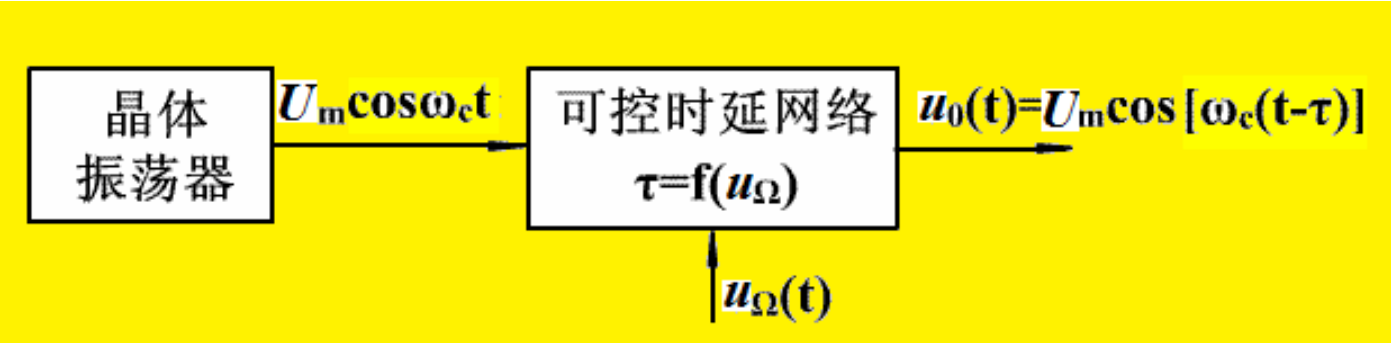
$$\Delta\omega_m = m_p\Omega = \frac{U_{\Omega m}\gamma Q_L}{R_3 C_4 (U_Q + U_D)}$$

上式表明最大频偏$\Delta\omega_m$与ω_c无关。

为了增加m_p,可采用多级单回路构成的变容管调相电路。



2. 可变时延法调相电路

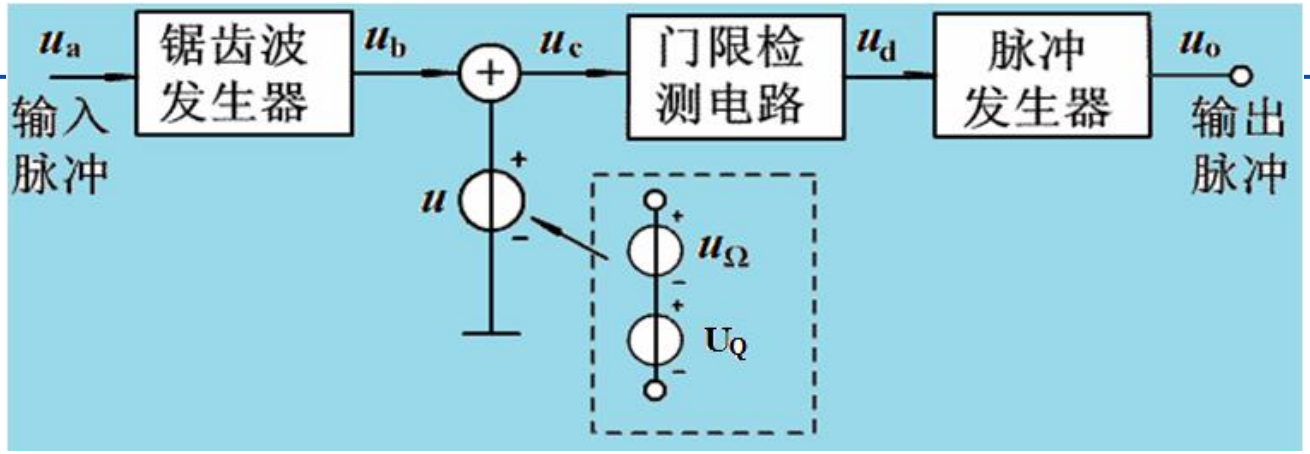


$$\tau = k_d u_\Omega = k_d U_{\Omega m} \cos \Omega t$$

则 $u_o(t)$ 为调相波：

$$u_0(t) = U_m \cos(\omega_c t - \omega_c k_d u_\Omega) = U_m \cos(\omega_c t - m_p \cos \Omega t)$$

$$m_p = \omega_c k_d U_{\Omega m}$$



单音调制时，$u_0(t) = U_m \cos[n\omega_c(t-\tau)]$

$m_p = n\omega_c \Delta\omega_m$　　$\Delta\omega_m$为最大时延差，n为谐波次数

理论上$\Delta\omega_m$=0. 5T_c，而实际上限制$\Delta\omega_m \le$0. 4T_c 。

$$\therefore m_p \le n\frac{2\pi}{T_c} \times 0.4T_c = 0.8n\pi$$

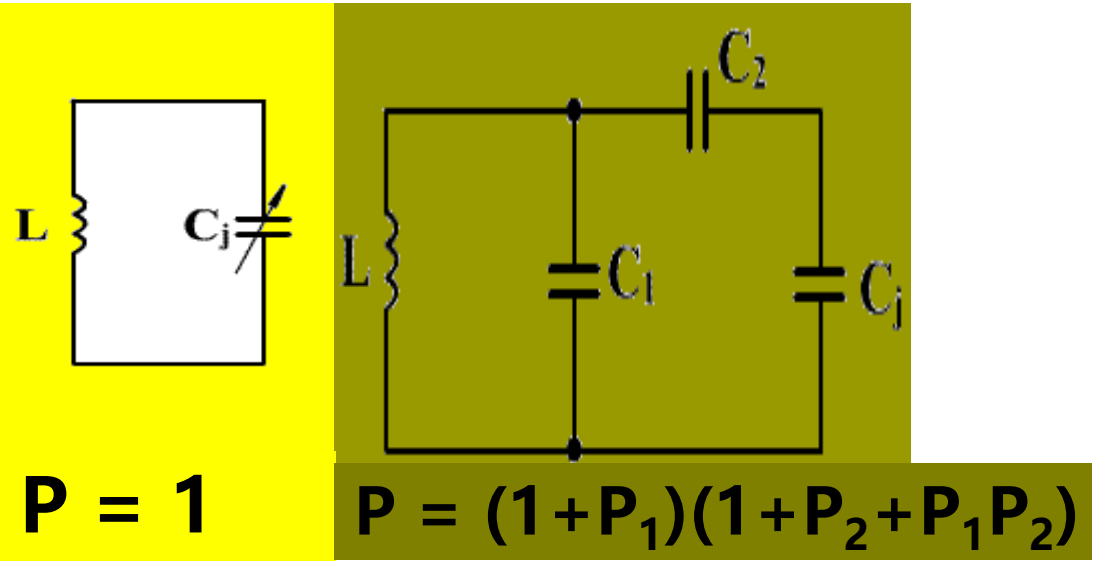
当$n = 1$时，$m_p = 0.8\pi$(2. 51rad或144°)　6-9.exe

可见，脉冲调相电路有线性相移较大的优点，用于调频广播发射机。

三、间接调频与直接调频电路在性能上的差别

1、直接调频

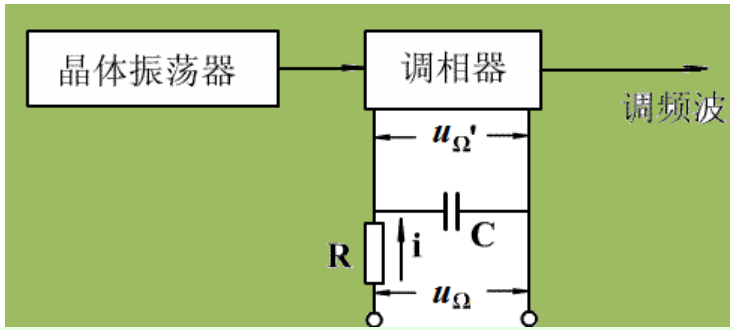
$$\Delta\omega_m = \frac{\gamma}{2P} M\omega_c$$



在直接调频电路中，最大频偏Δ ω_m与ω_c 成正比，

而在间接调频电路中，最大频偏Δ ω_m与ω_c 无关。

2、间接调频



$$\Delta\omega_m = m_p\Omega = \frac{U_{\Omega m}\gamma Q_L}{RC\ (U_Q + U_D)}$$，与ω_c无关；

$$m_f = m_p = \frac{U_{\Omega m}Q_L\gamma}{\Omega RC(U_D + U_Q)}$$，与ω_c无关；与Ω成反比。

如果调制信号中包含Ω_{min}、…、Ω_{max}众多频率分量，则当$\Delta\omega_m$一定时（$U_{\Omega m}$一定时）,Ω越小,m_f就越大。

当$\Omega = \Omega_{min}$时，m_f达到最大值。

在间接调频电路中，最大频偏必发生在最低调制信号频率上，$\Delta\omega_m = m_f \Omega_{\min}$

例如 F : 100—3400Hz, 这时F_{min}=100Hz

当采用变容二极管调相时，

$$m_f = \frac{\pi}{6} = 0.52\text{rad}, \Delta f_m = 52\text{Hz}$$

当采用三级变容管调相时，

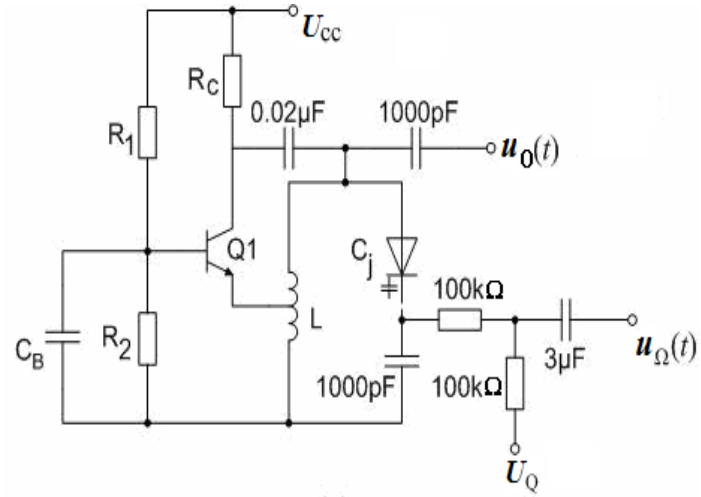
$$m_f = \frac{\pi}{2} = 1.57\text{rad}, \Delta f_m = 157\text{Hz}$$

当采用脉冲调相时，$m_f = 0.8\pi = 2.5\text{rad}, \Delta f_m = 250\text{Hz}$

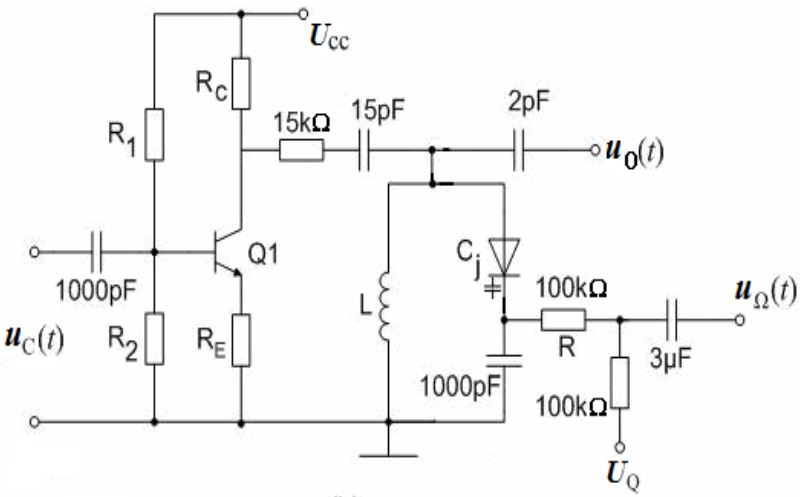
例：两个角度调制电路如图所示。$u_c(t)=2\cos10^7t$ (V)，$u_\Omega(t)=0.1\cos10^3t$ (V)，$U_Q=4.4$ (V)，变容管的参数为 $U_D=0.6$V，$\gamma=2$，$C_{jQ}=40$pF，回路品质$Q_L=100$， 指出各个电路输出的电压$u_0(t)$为调频波还是调相波，并分别计算

(1)最大角频偏$\Delta\omega_m$； (2)输出瞬时角频率$\omega(t)$；

(3)调制频率增加1倍时的$\Delta\omega_m$；(4)载频增加1倍时的$\Delta\omega_m$。



(a)



(b)

解：从题所示电路结构上可知(a)图为调频波电路，

(b)图为调相电路，

再分析：，$\Omega RC = 10^3 \times 100 \times 10^3 \times 1000 \times 10^{-12} = 0.1$

因此(b)图输出为PM波。

对(a)图属于变容管全部接入方式的直接调频，

LC_j回路与晶体管构成电感三点式振荡器。

对(a)图

(1) $$M=\frac{U_{\Omega m}}{U_D+U_Q}=\frac{0.1}{0.6+4.4}=0.02$$

$$\Delta\omega_m=\frac{\gamma}{2}M\omega_c=\frac{2}{2}\times 0.02\times 10^7=2\times 10^5(\text{rad/s})$$

(2) $$\omega(t)=10^7+2\times 10^5\cos 10^3 t \quad (\text{rad/s})$$

(3) $\Omega\uparrow$ $\Delta\omega_m$不变 $\Delta\omega_m=2\times 10^5(\text{rad/s})$

(4) $\omega_c\uparrow$ $\Delta\omega_m=4\times 10^5(\text{rad/s})$

对(b)图，变容管作为全接入方式时P=1

(1) 调相指数 $m_p = \mathrm{M}\gamma Q_L = 0.02\times 2\times 100 = 4\ \mathrm{rad} > \dfrac{\pi}{6}$

而实际上 $m_p' \le \dfrac{\pi}{6} = 0.523$

$$\therefore \Delta\omega_m = \frac{\pi}{6}\times\Omega = 0.523\times 10^3 = 523(\mathrm{rad/s})$$

(2) $$\varphi(t) = \omega_c t + m'_p \cos 10^3 t$$

$$\omega(t) = \frac{d\varphi(t)}{dt} = \omega_c - m_p'\Omega\sin 10^3 t = 10^7 - 523\sin 10^3 t(\mathrm{rad/s})$$

(3) $\Omega\uparrow$ m_p不变 $\Delta\omega_m = 2\times 523 = 1046(\mathrm{rad/s})$

(4) $\omega_c\uparrow$ $\Delta\omega_m = 523(\mathrm{rad/s})$不变

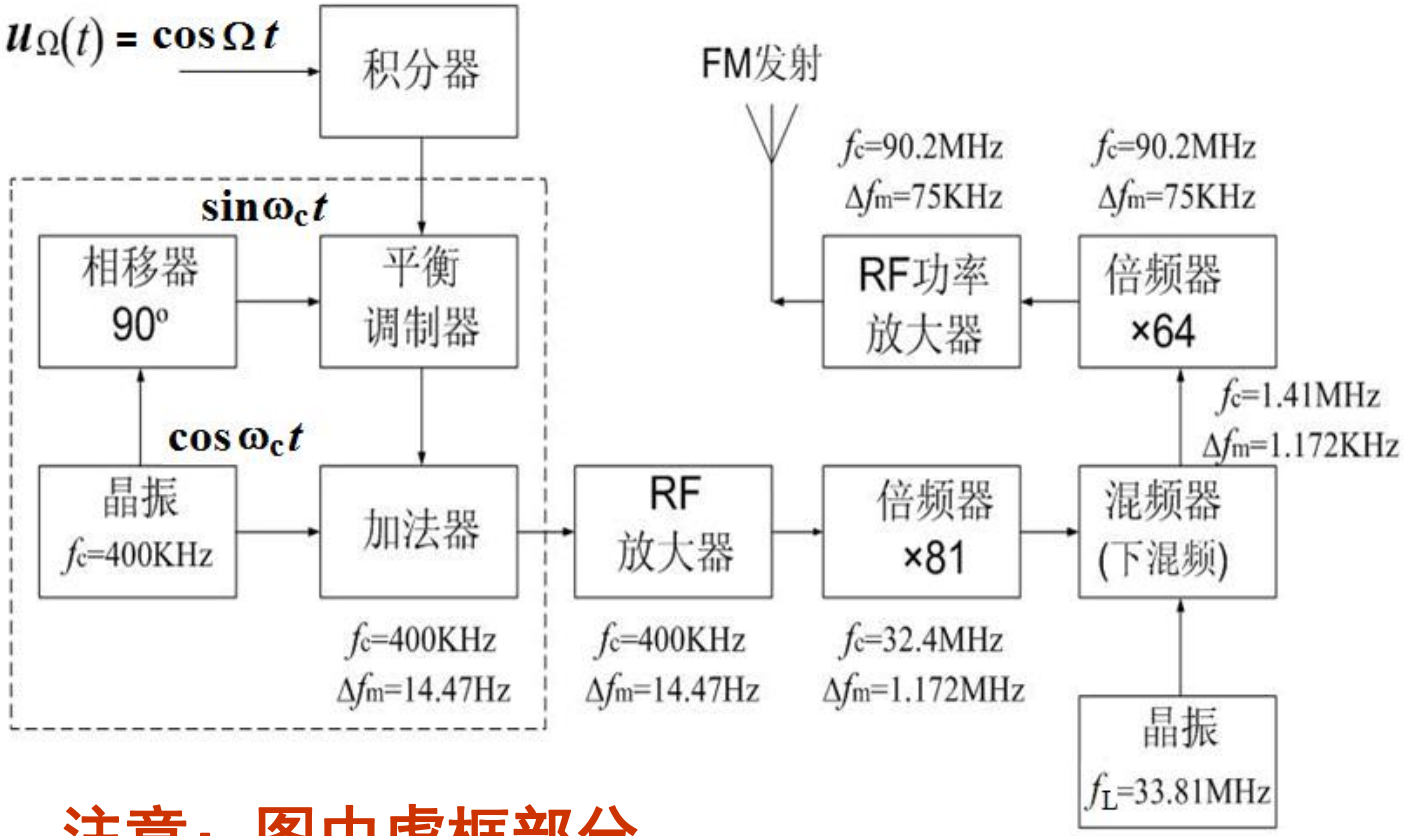
四、扩展线性频偏的方法

方法1: 倍频

6-10.exe

方法2：混频 + 倍频

6-11.exe



注意：图中虚框部分

Armstrong间接调频发射机

当$u_{\Omega 2}(t)=U_{\Omega m}\sin\Omega t$时

$$u_{\mathrm{PM}}(t)=U_m\cos(\omega_c t+m_p\sin\Omega t)$$

$$=U_m[\cos\omega_c t\cos(m_p\sin\Omega t)-\sin\omega_c t\sin(m_p\sin\Omega t)]$$

若m_p<<1，则cos ($m_p$sinΩt)≈1，sin($m_p$sinΩt)≈$m_p$sinΩt，

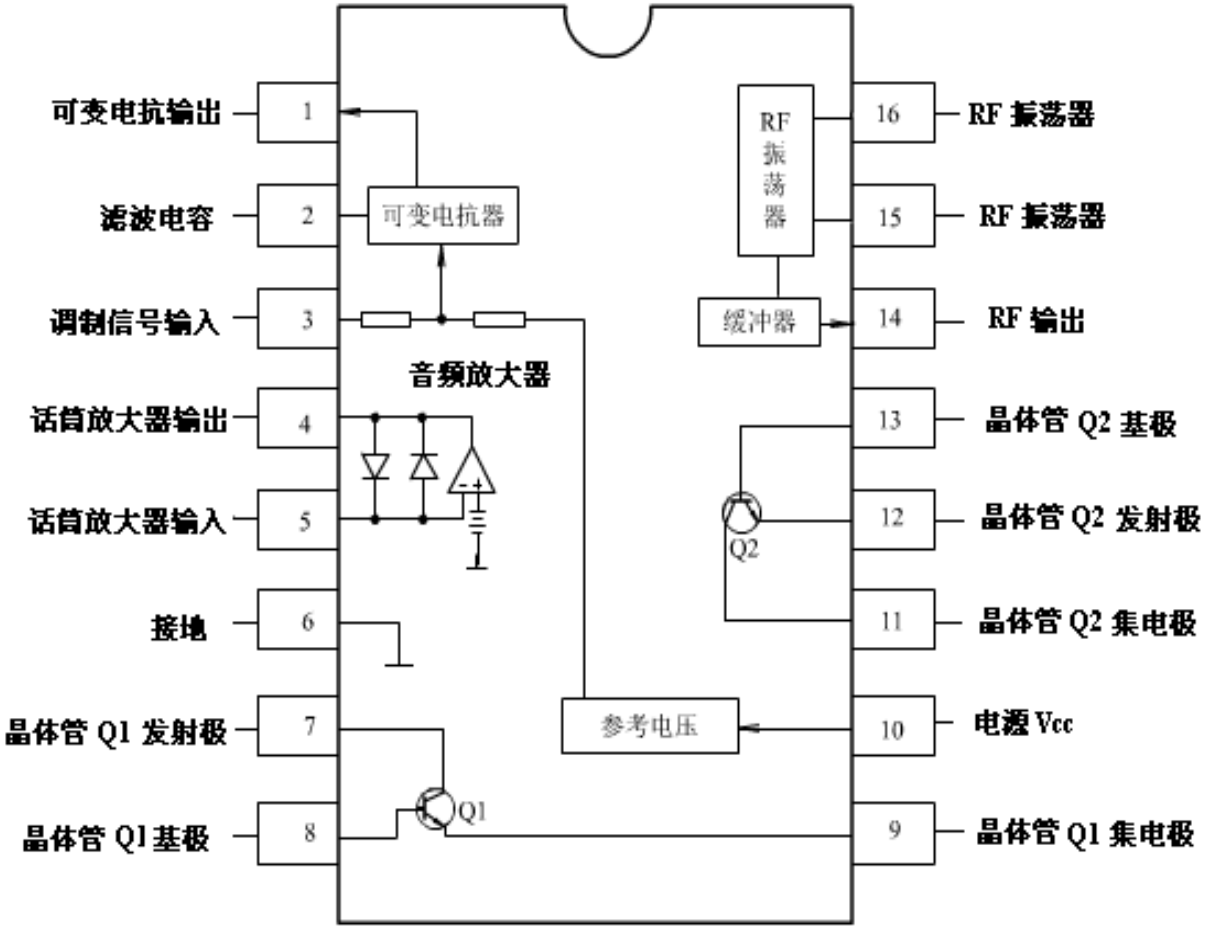
这时，有

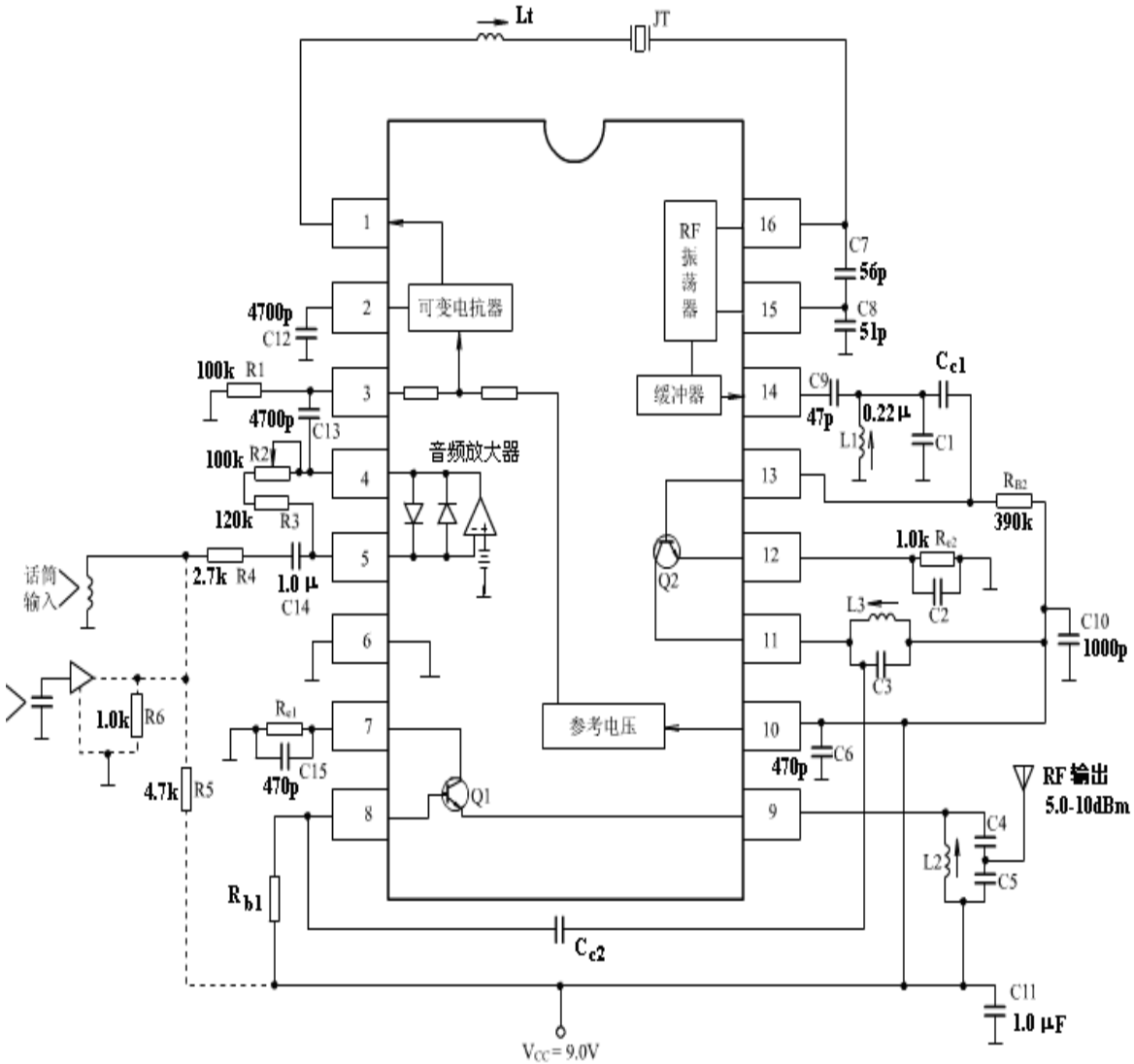
$$u_{\mathrm{PM}}(t)=U_m\,[\cos\omega_c t-m_p\sin\Omega t\sin\omega_c t]$$

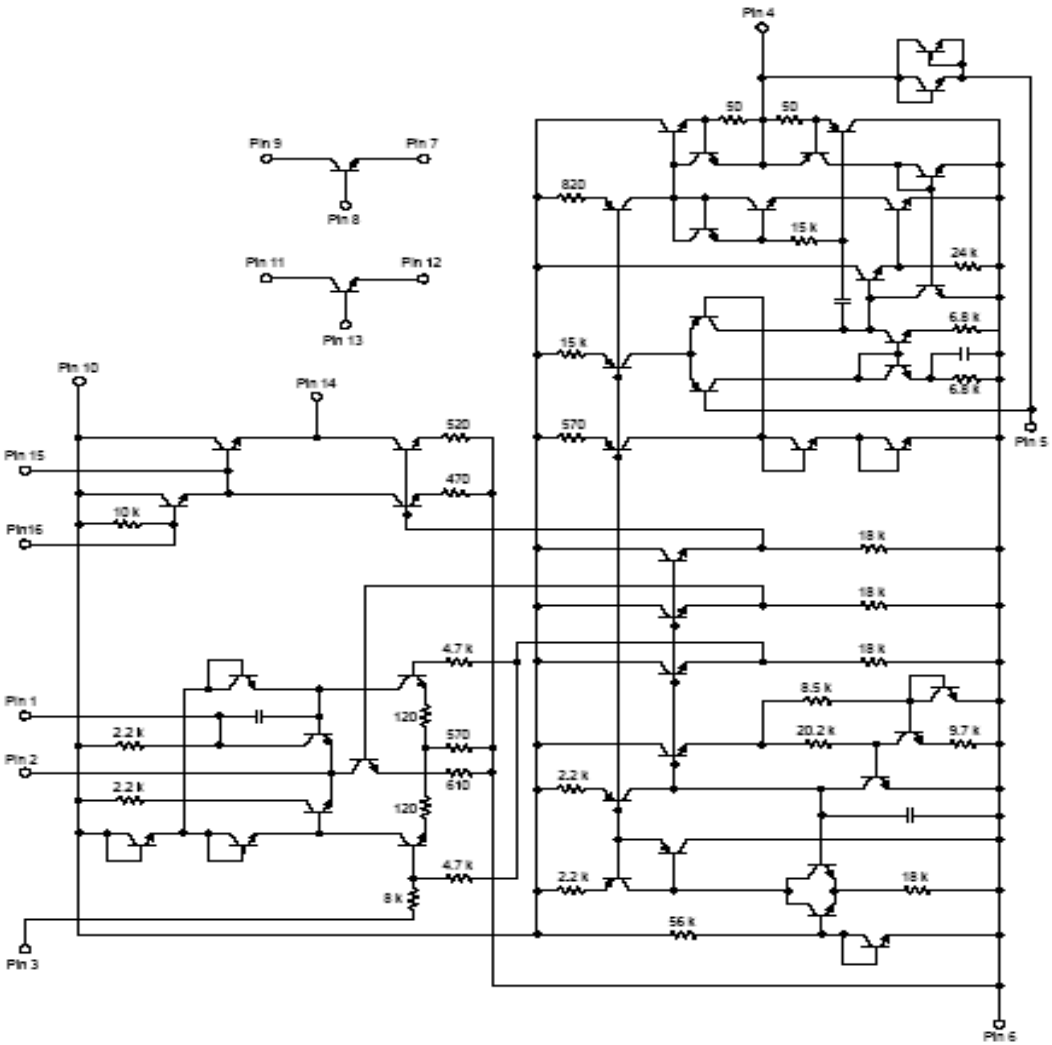
小调相指数(亦即小频偏)的调相波可由载波和一个与载波正交的双边带信号叠加而成。

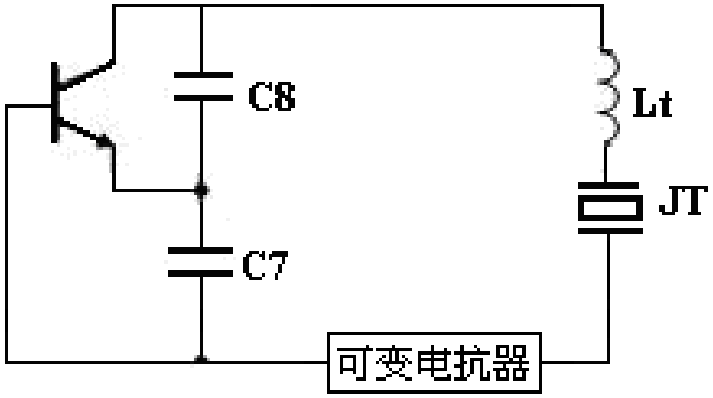
五、集成电路MC2833

MC2833是单片调频发射集成电路




Lt
JT
RF
振
荡
器
可变电抗器
缓冲器
音频放大器
参考电压
C7
56p
C8
51p
4700p
C12
100k R1
4700p
C13
100k R2
120k R3
2.7k R4
1.0 μ
C14
话筒
输入
1.0k R6
4.7k R5
Re1
C15
470p
Rb1
Q1
Q2
C9
47p
0.22 μ
L1
Cc1
C1
RB2
390k
1.0k Re2
C2
L3
C3
C10
1000p
C6
470p
RF 输出
5.0-10dBm
C4
L2
C5
Cc2
C11
1.0 μF
VCC = 9.0V


Pin 4
50
50
Pin 9
Pin 7
Pin 8
820
15 k
Pin 11
Pin 12
Pin 13
24 k
6.8 k
15 k
6.8 k
Pin 10
Pin 14
520
570
Pin 5
Pin 15
470
10 k
Pin16
18 k
18 k
18 k
4.7 k
Pin 1
120
8.5 k
2.2 k
570
20.2 k
9.7 k
Pin 2
2.2 k
610
2.2 k
120
4.7 k
2.2 k
18 k
8 k
56 k
Pin 3
Pin 6



振荡器和调频电路合二为一，均由晶体振荡器完成：

§6.4 限幅器

调角波在产生过程中或多或少附带寄生调幅，当传输到接收端时，也会在调频波上出现寄生调幅。通常用限幅器来消除调频波的寄生调幅。

限幅器的组成： **非线性元件 + 谐振回路**

常用限幅器： **晶体二极管限幅器**

晶体三极管限幅器

差分对限幅器

§6.5 鉴频器

鉴频方法：

1、**斜率鉴频器，相位鉴频器，比例鉴频器**

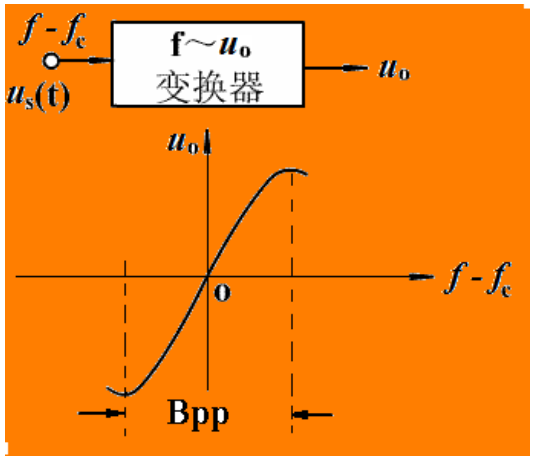
2、**移相正交鉴频器**

3、**脉冲计数式鉴频器**

6-12.exe

一. 主要指标

鉴频特性曲线：



1. 灵敏度 S_D ——

f_c处的鉴频特性曲线的斜率

$$S_D = \left.\frac{du_o}{df}\right|_{f=f_c}$$ (要求大)

2. 线性范围： —— $B_{PP} \geq 2\Delta f_m$ (要求大)

3. 非线性失真 (要求小)

二. 斜率鉴频器

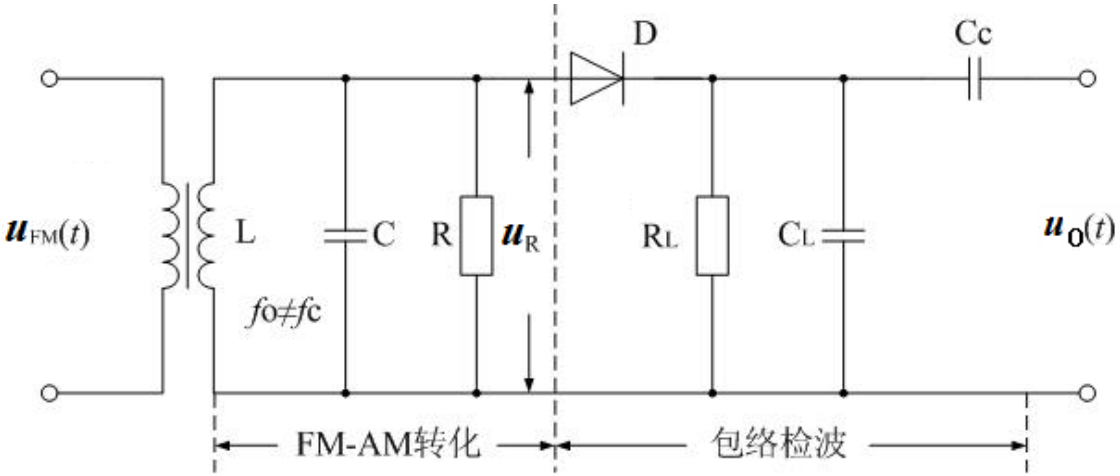
实现框图：

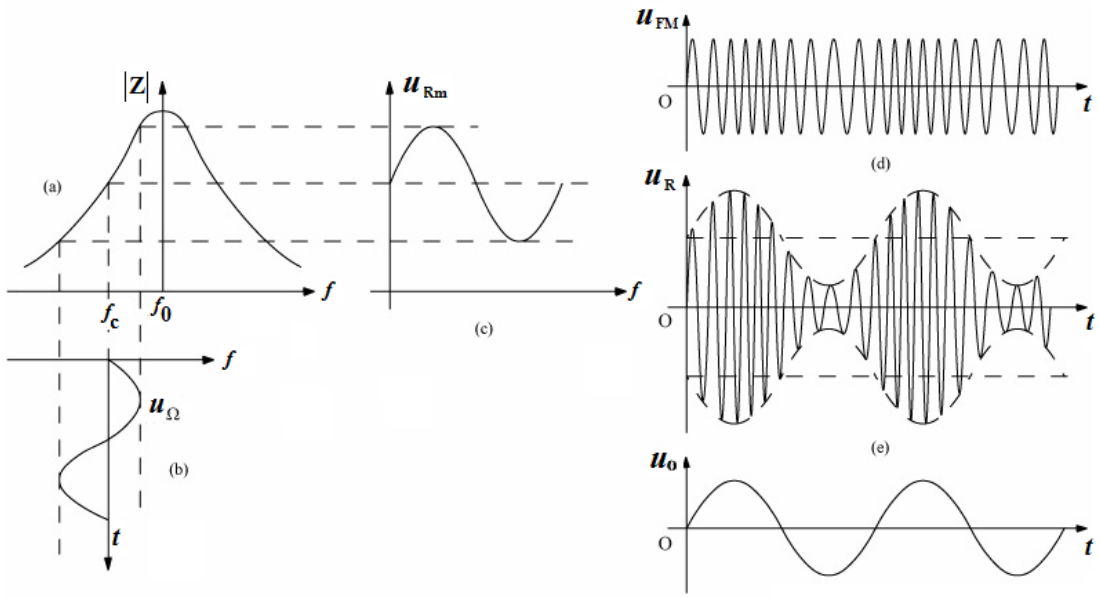
FM → 线性网络 → 包络检波器 → $u_\Omega(t)$

(变换器)

FM → AM

1. 单失谐的斜率鉴频器





输入FM波：

$$u_{\mathrm{FM}}(t) = U_m \cos(\omega_c t + m_f \sin \Omega t)$$

$u_R(t)$变为幅度反映瞬时频率变化的调频波，称其为调频-调幅(FM-AM)波。而后通过包络检波器完成鉴频功能。

$$U_{Rm} = I_m |Z|$$

$$|Z| = \frac{R_p}{\sqrt{1 + Q_L^{\,2}\left(\frac{f}{f_0} - \frac{f_0}{f}\right)^2}} = \frac{R_p}{\sqrt{1+\xi^2}}$$

$$\therefore u_0 = K_d U_{Rm} = K_d I_m |Z|$$

$$S_D = \frac{du_0}{df} = \frac{d\left(k_d I_m |Z|\right)}{df} = K_d I_m \frac{d|Z|}{df}$$

$$= K_d I_m R_p \frac{d}{d\xi}\left(\frac{1}{\sqrt{1+\xi^2}}\right)\frac{d\xi}{df}$$

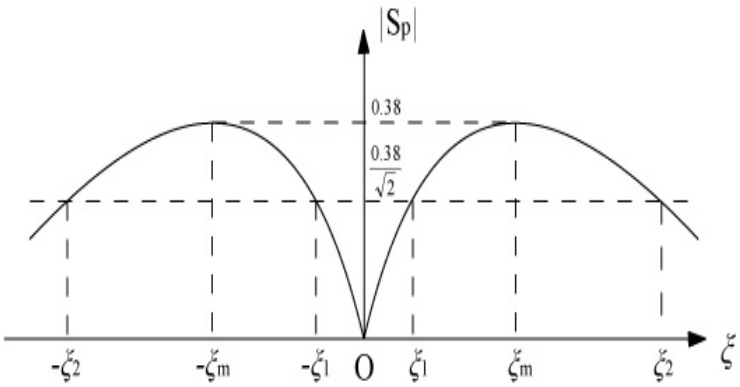
$$\xi \approx Q_L \frac{2(f - f_0)}{f_0} \qquad \frac{d\xi}{df} = \frac{2Q_L}{f_0}$$

$$S_D = 2k_d I_m R_p \cdot \frac{Q_L}{f_0} \cdot \frac{d}{d\xi}\left(\frac{1}{\sqrt{1+\xi^2}}\right)$$

$$= \frac{2k_d I_m R_p Q_L S_p}{f_0}$$

其中 $S_p = \frac{d}{d\xi}\left(\frac{1}{\sqrt{1+\xi^2}}\right)$

$$|S_p| = \frac{d}{d\xi}\left(\frac{1}{\sqrt{1+\xi^2}}\right) = \frac{|\xi|}{(1+\xi^2)^{3/2}}$$



在 $\xi_m = \pm 0.7$ **处，** $|S_p|$ **取得最大值，即** $|S_p|_{\max} \approx 0.38$

可得最大鉴频灵敏度

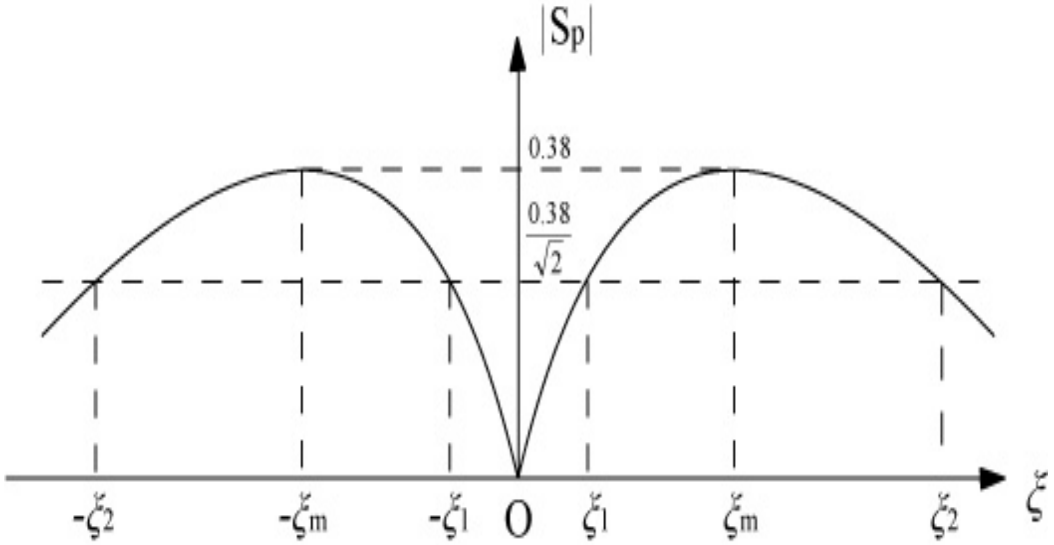
$$\boxed{|S_D|_{\xi_m=0.7} \approx \frac{0.76 K_d I_m R_p Q_L}{f_0}}$$

取较小的对称区域作为频偏，取

$$\Delta\xi = \xi_m - \xi_1 \approx 0.4$$

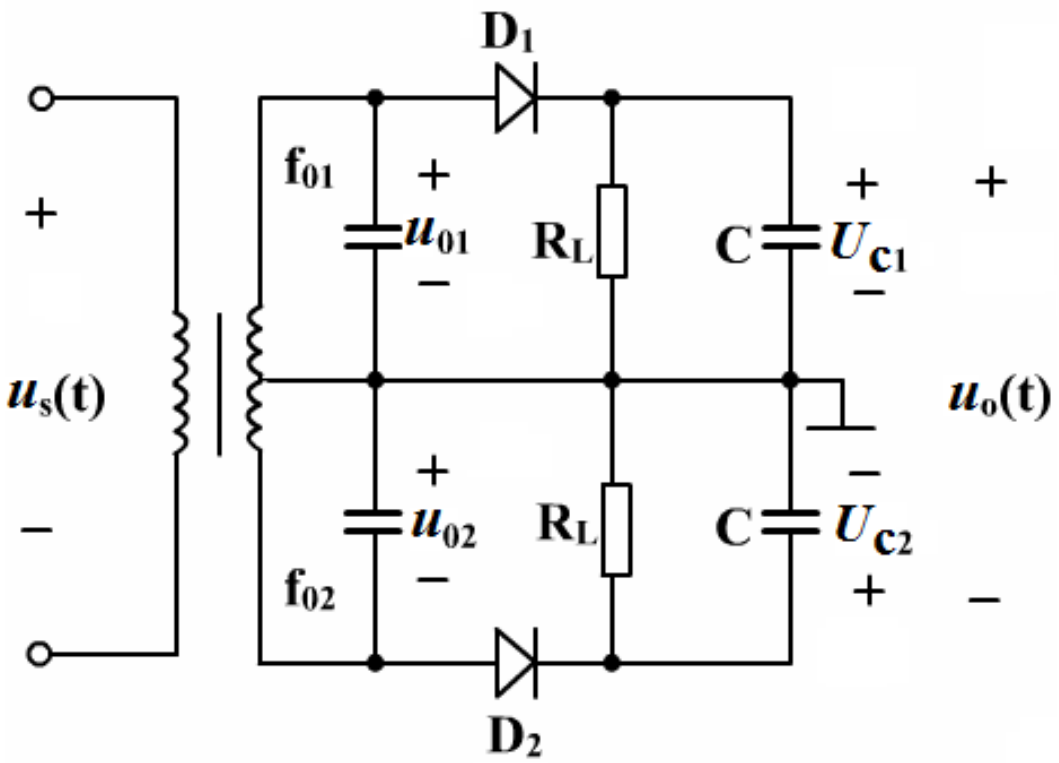
$$\xi = 2Q_L \frac{\Delta f}{f_0}$$

$$\Delta f \approx 0.2 \frac{f_0}{Q_L}$$



$$\therefore B_{pp} = 2 f_{\max} \le 2\Delta f = 0.4 \frac{f_0}{Q_L}$$

2. 平衡斜率鉴频器



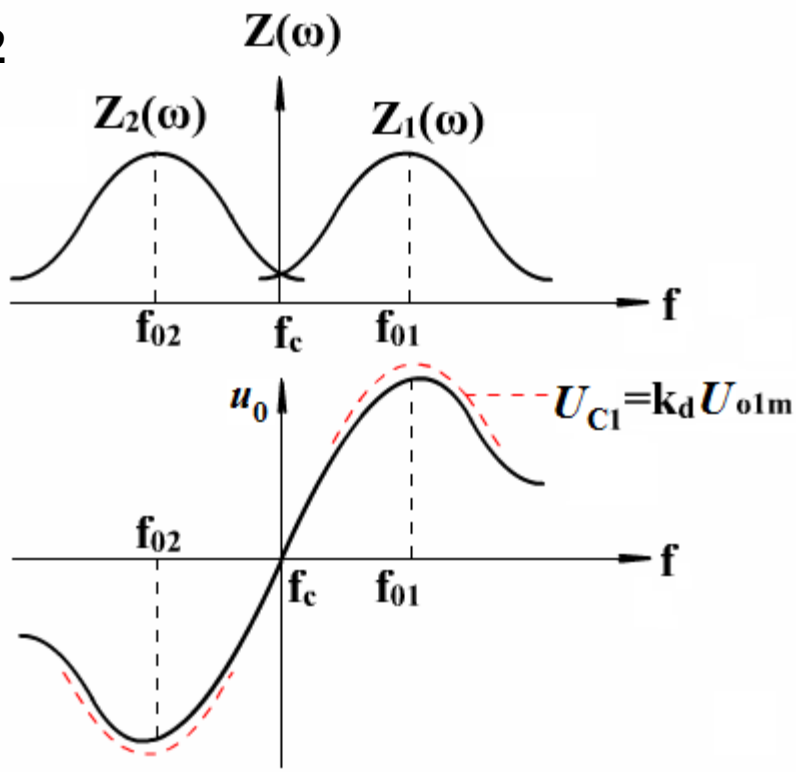
实际上，单调谐回路的谐振曲线，其倾斜部分的线性范围是很小的，为了扩展鉴频特性的线性范围，实用的斜率鉴频器都是采用两个单失谐回路构成的平衡电路。

成对称失谐：$f_{01} - f_C = f_C - f_{02}$

设包络检波效率均为k_d则:

$$u_0 = U_{C1} - U_{C2}$$

$$= k_d (U_{01m} - U_{02m})$$



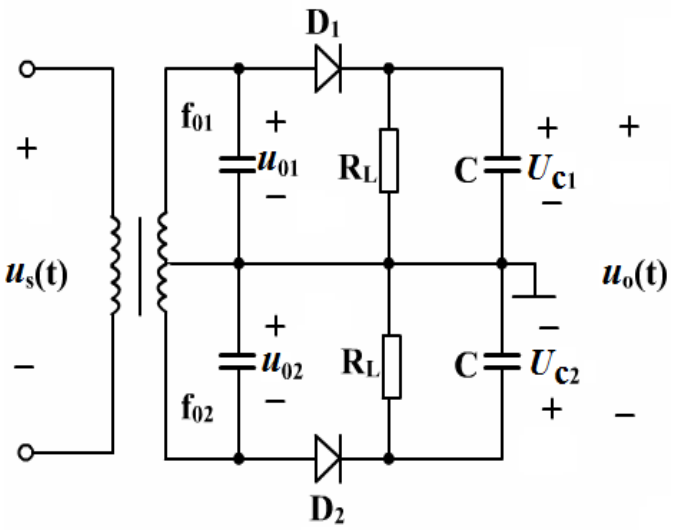
Δf 不能过大，也不能过小，

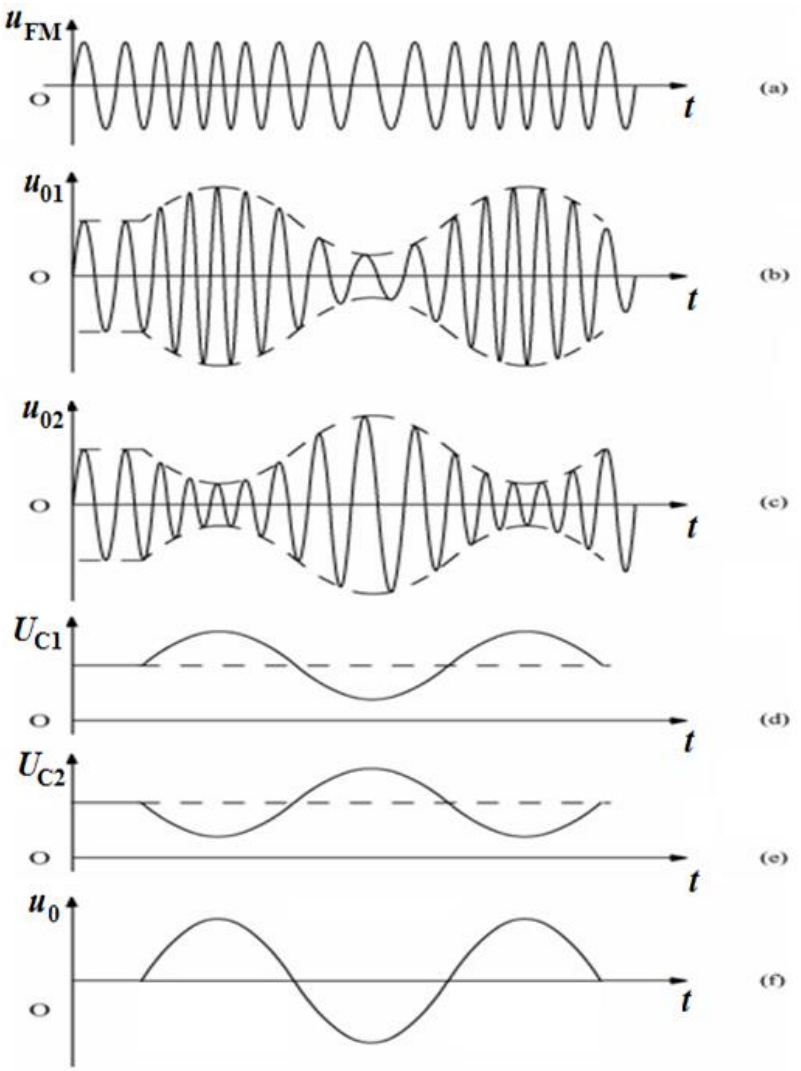
$$B_{PP}' = 2f'_{\max} \le 2(f_{01} - f_c) = 1.1\frac{f_{01}}{Q_L}$$

$$S_D' \approx \sqrt{2} S_D$$

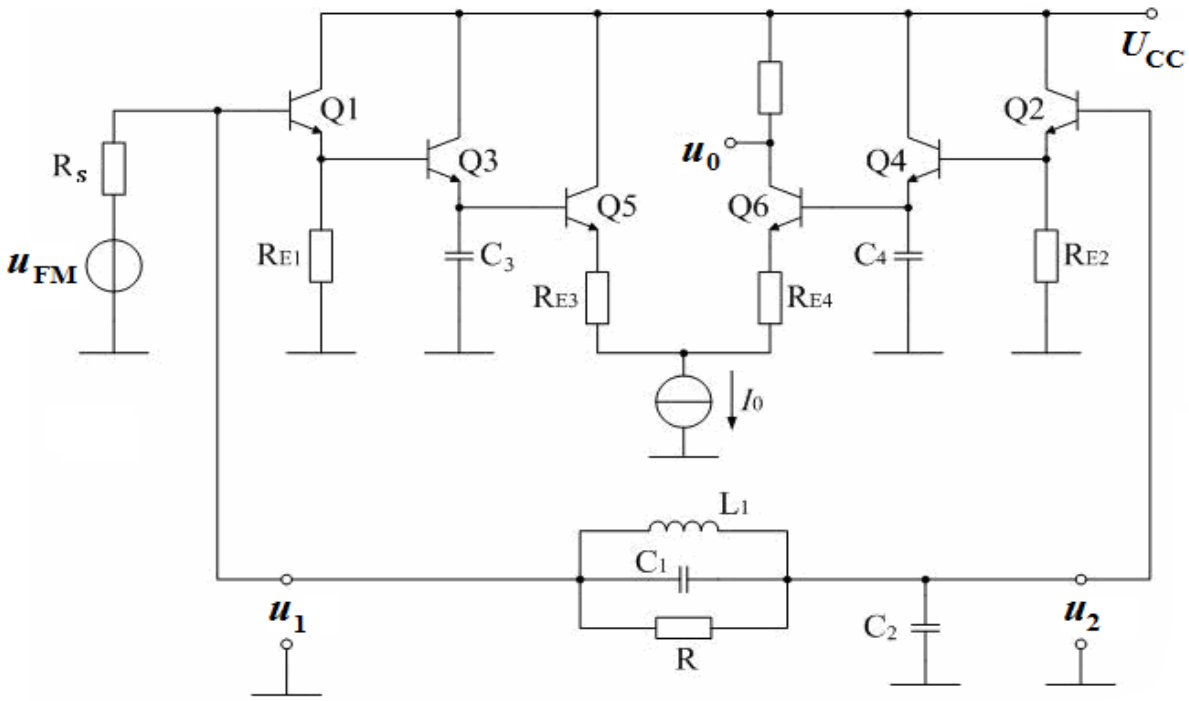
则鉴频特性的线性范围达到最大

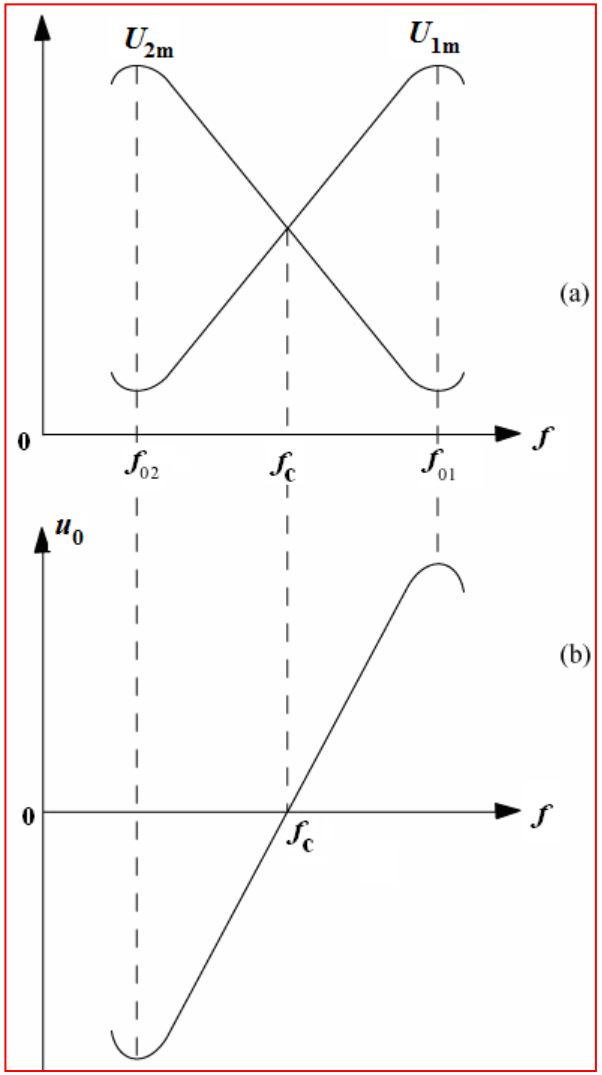
平衡斜率鉴频器
主要工作波形图





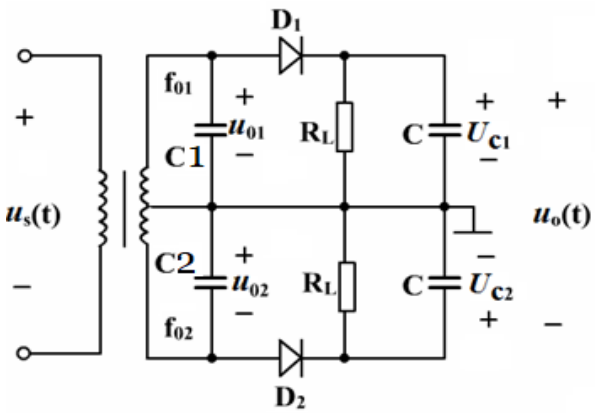
3.集成电路中采用的斜率鉴频器



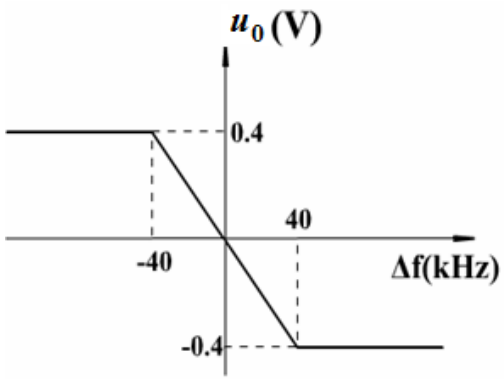


差分峰值鉴频器

例：互感耦合相位鉴频器电路如题图所示，



图（a）



图(b)

(1) 若$f_{01}<f_c$，$f_{02}>f_c$，且$f_c-f_{01}=f_{02}-f_c$，分析其工作原理；

(2)画出电容C1两端的电压波形示意图；

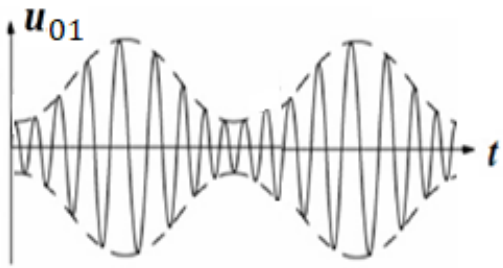
(3)若鉴频特性如题图(b)所示，S_D=-10mV/kHz，

$u_s=1.5\cos(2\pi\times10^7t+15\sin4\pi\times10^3t)$(V)，求输出$u_o$=？

(4)当发送端调制信号$U_{\Omega m}$加大一倍时，画出$\boldsymbol{u}_o$的波形图。

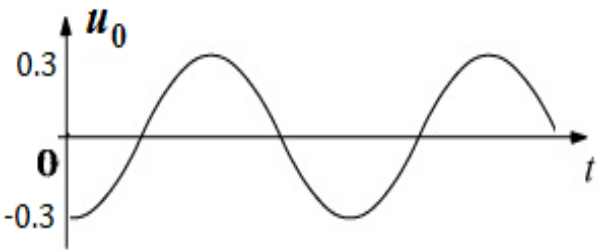
解：

（2）



(3)由于$u_s = 1.5\cos(2\pi\times10^7 t + 15\sin 4\pi\times10^3 t)$

$\Delta\varphi(t) = 15\sin 4\pi\times10^3 t$（rad/s)



$$\Delta f(t) = \frac{1}{2\pi}\frac{d\Delta\varphi(t)}{dt} = 2\times10^3\times15\cos 4\pi\times10^3 t$$

$$= 30\cos 4\pi\times10^3 t(\text{kHz}),$$

$$\Delta f_m = 30\text{kHz}, \because S_D = 10\text{mV/kHz}, B_{pp} = 80\text{kHz},$$

$$\therefore u_0 = S_D\times\Delta f(t) = -10\times10^{-3}\times30\cos 4\pi\times10^3 t = -0.3\cos 4\pi\times10^3 t(\text{V})$$

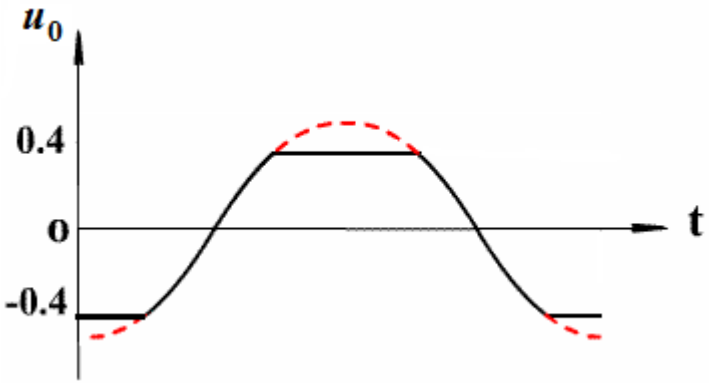
(4)$U_{\Omega m}$加大一倍，则Δf_m也加大一倍，

即$\Delta f_m^{'} = 2\times 30\text{kHz}$,

$$\Delta f_m^{'} = 60\text{kHz} > \frac{B_{pp}}{2},$$

$$u_0 = S_D \Delta f(t) = \begin{cases} -0.6\cos(4\pi\times 10^3 t)(\text{V}) & |\Delta f(t)| < 40\text{kHz} \\ 0.4(\text{V}) & \Delta f(t) < -40\text{kHz} \\ -0.4\text{V} & \Delta f(t) > 40\text{kHz} \end{cases}$$

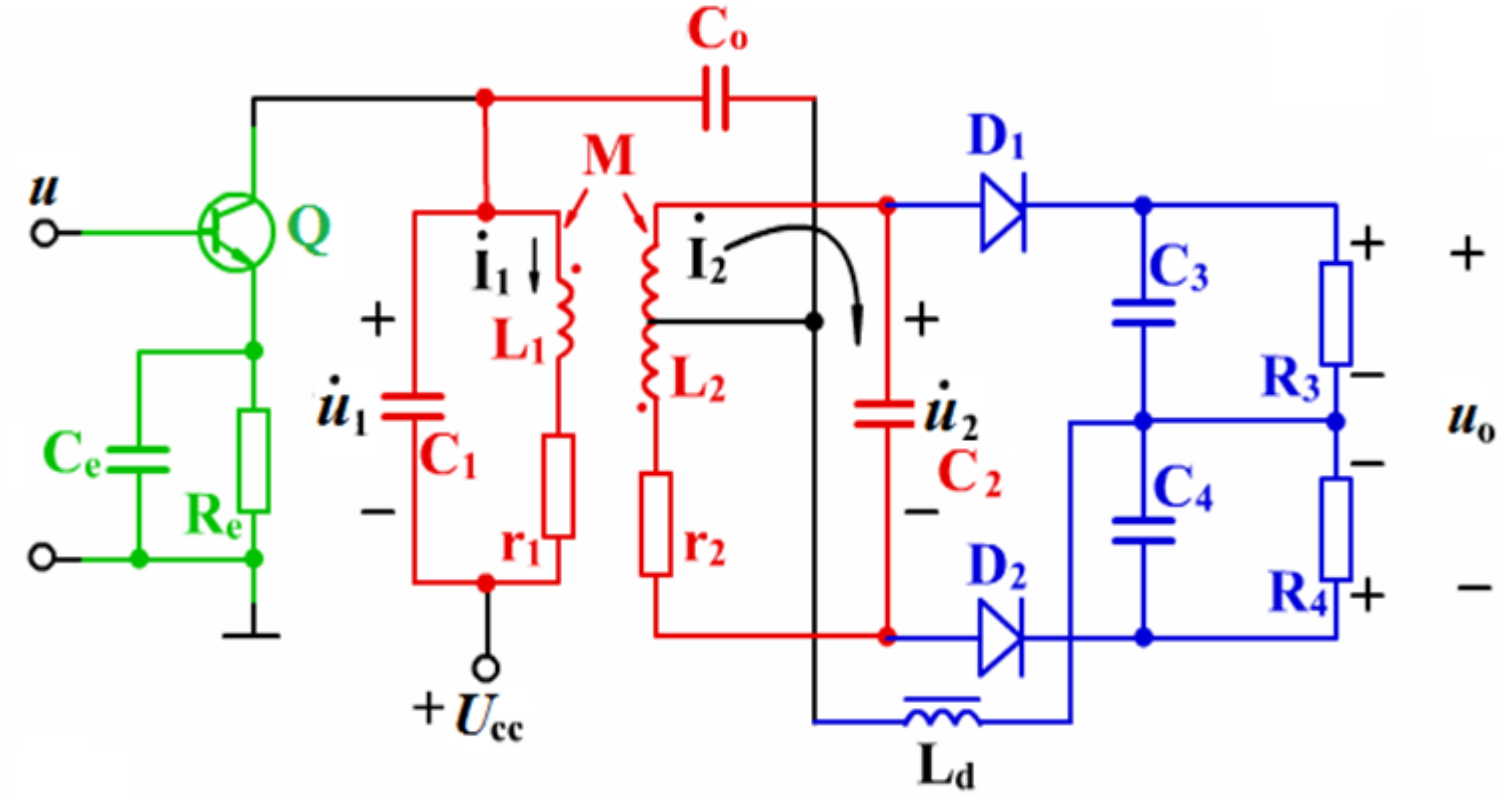
在接收端产生失真。



三. 相位鉴频器

相位鉴频器是利用耦合谐振回路次级和初级间的相移随频率变化的特性来实现鉴频的。

下面介绍利用互感耦合相位鉴频器的原理。



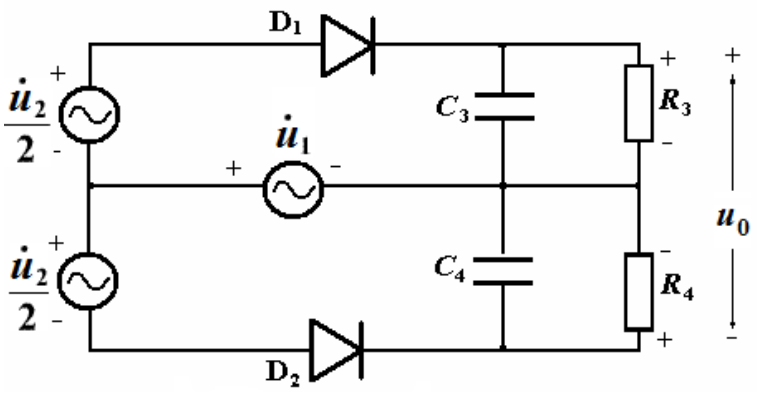
1、作为移相网络，初级次级都调谐于f

2. 平衡型叠加鉴相器

6-13.exe

3、C_0在耦合时，对$\dot{u}_1$短路，$\dot{u}_1$经C_0全部降在L_d上

4、$\dot{u}_1$通过M到次级电压$\dot{u}_2$，中心对称$\dfrac{\dot{u}_2}{2}$



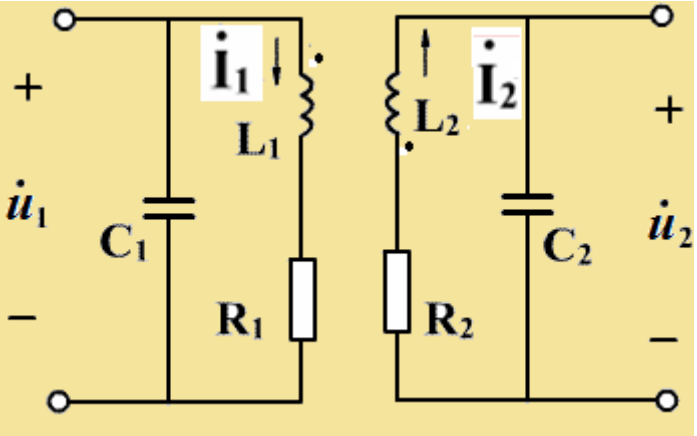
(b)相位鉴频器检波回路等效电路

(b)图中，加在二级管D_1和D_2两端的高频电压为：

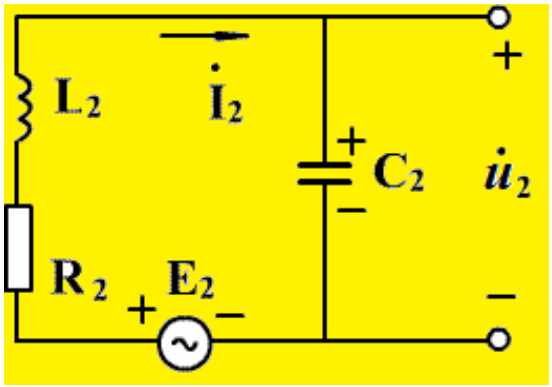
$$\dot{u}_{D1} = \dot{u}_1 + \frac{\dot{u}_2}{2}$$

$$\dot{u}_{D2} = \dot{u}_1 - \frac{\dot{u}_2}{2}$$

1. 频相转换网络



(a)初、次级回路



（b）次级回路

假设，图(a)中，初次级回路参数完全相同，即 $L_1 = L_2$

$R_1 = R_2, Q_1 = Q_2, \xi_1 = \xi_2 = \xi$，则得初、次级回路的阻抗为：

$$Z_2 = R_2 + j\omega L_2 + \frac{1}{j\omega c_2} \approx R_2(1 + jQ_L \frac{2\Delta\omega}{\omega_0}) = R(1 + j\xi)$$

初级回路$\dot{I}_{L1}$通过互感在次级产生的感应电势为：

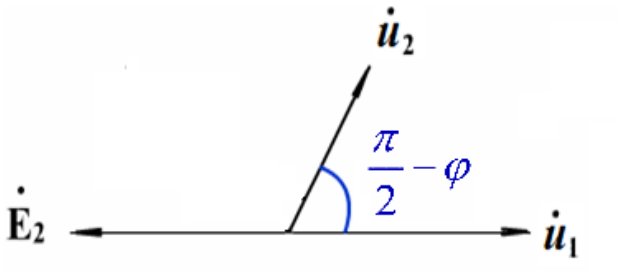
$$\dot{E}_2 = -j\omega M\dot{I}_1 = -j\omega M\frac{\dot{u}_1}{j\omega L_1} = -k\dot{u}_1 \qquad k = \frac{M}{L}$$

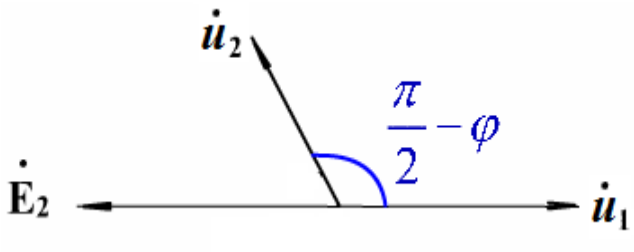
次级回路电流$\dot{I}_2$在电容C_2两端产生的电压为：

$$\dot{u}_2 = \frac{\dot{I}_2}{j\omega C_2} = \frac{\dot{E}_2}{j\omega C_2 Z_2} = \frac{jkQ_L\dot{u}_1}{1+j\xi} = j\frac{\eta\dot{u}_1}{1+j\xi} = \frac{\eta\dot{u}_1}{\sqrt{1+\xi^2}}\angle\frac{\pi}{2}-\varphi$$

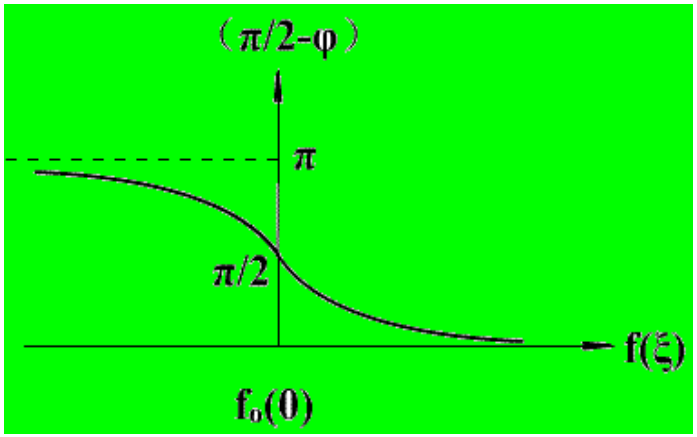
式中，$\xi = Q_L\dfrac{2\Delta\omega}{\omega_0}$,$\xi$为回路广义失谐，$\varphi = \arctan\xi$

$$k = \frac{M}{\sqrt{L_1L_2}} = \frac{M}{L}$$,耦合系数，$\eta = kQ_L$,耦合因数

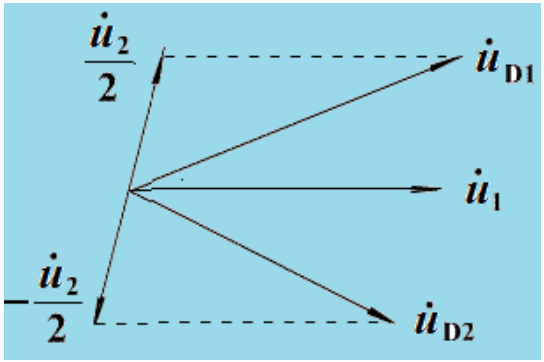




在上式中，广义失谐ξ是与信号频率有关的量。所以当调频波的频率变化时，相对初级回路端压$\dot{u}_1$和次级回路$\dot{u}_2$的幅度和相位将随之而变化，下面画出随着f变化的$\dot{u}_1$和$\dot{u}_2$的相位差特性。

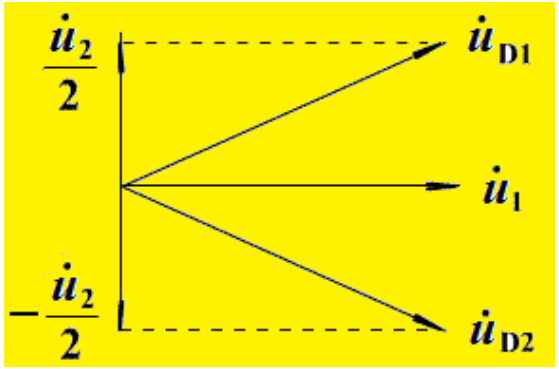


2. 矢量图：



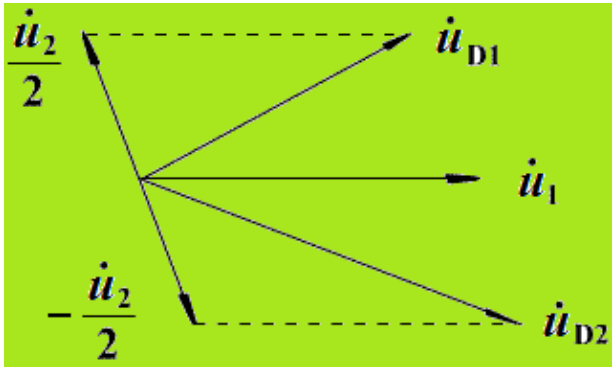
$f > f_C$

$u_O > 0$



$f = f_C$

$u_O = 0$



$f < f_C$

$u_O < 0$

当$\omega > \omega_0, \xi > 0, \dot{u}_1$和$\dot{u}_2$相位相差$< 90^o$.

于是$|\dot{u}_{D1}|\uparrow, |\dot{u}_{D2}|\downarrow$, 故$u_0 > 0$

当$\xi \to \infty$， $u_0 = k_d(|\dot{u}_{D1}| - |\dot{u}_{D2}|)$

$$= k_d[(U_{1m} + \frac{U_{2m}}{2}) - (U_{1m} - \frac{U_{2m}}{2})]$$

$$= k_d U_{2m}$$

当$\omega < \omega_0, \xi < 0, \dot{u}_1$和$\dot{u}_2$相位相差$> 90^o$.

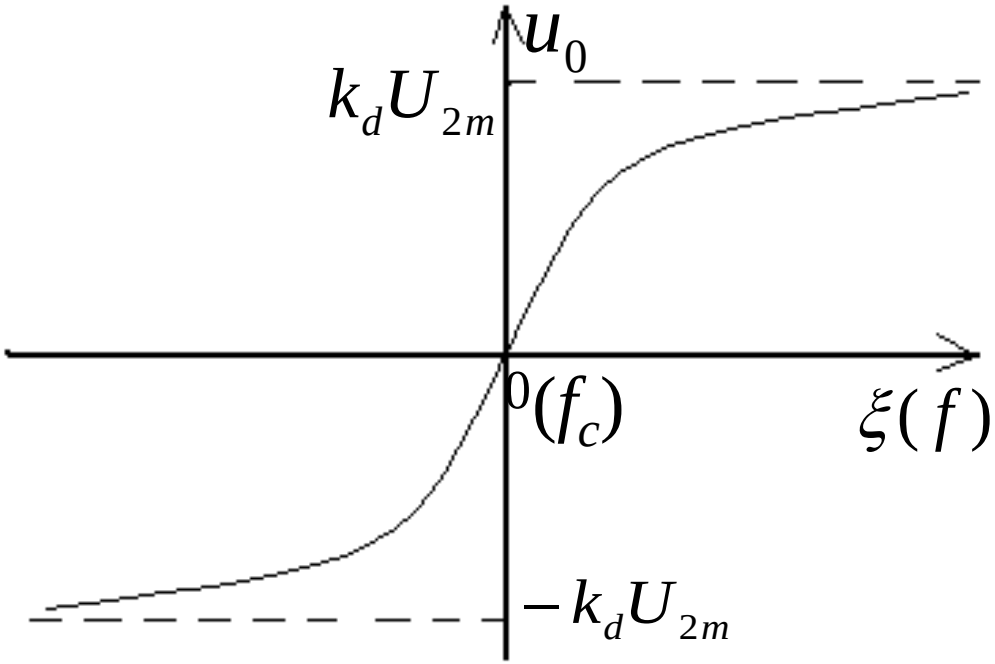
于是$|\dot{u}_{D1}|\downarrow, |\dot{u}_{D2}|\uparrow$, 故$u_0 < 0$

当$\xi \to -\infty$， $u_0 = k_d(|\dot{u}_{D1}| - |\dot{u}_{D2}|)$

$$= k_d[(U_{1m} - \frac{U_{2m}}{2}) - (U_{1m} + \frac{U_{2m}}{2})]$$

$$= -k_d U_{2m}$$

综上所述，当调频信号的瞬时频率以ω_c为中心而按调制信号的规律变化时，ξ以及$\dot{u}_1$和$\dot{u}_2$间的相位差，$|\dot{u}_{D1}|$和$|\dot{u}_{D2}|$的幅值等，都随之变化，从而形成了调幅调频波。经过检波后的输出电压即为调制电压。



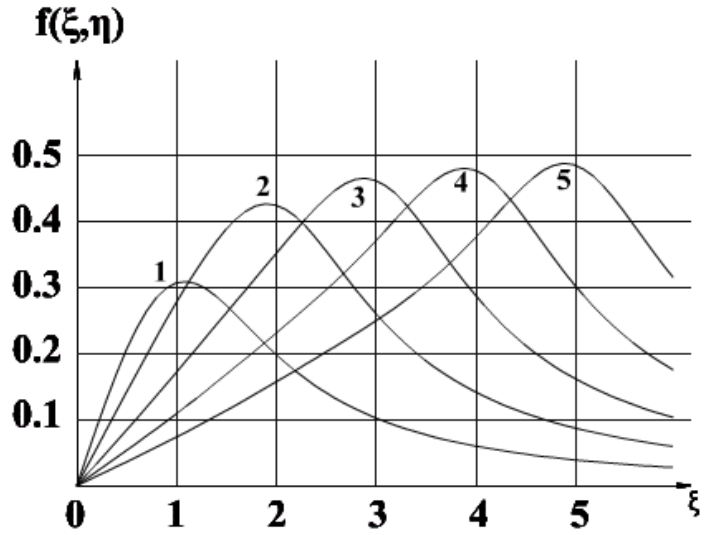
对u_0作精确推导：

$$u_0 = K_d\left(\left|u_{D1}\right| - \left|u_{D2}\right|\right)$$

$$= K_d I_1 R_e f(\xi,\eta)$$

$$f(\xi,\eta) = \frac{\sqrt{4+(2\xi+\eta)^2} - \sqrt{4+(2\xi-\eta)^2}}{2\sqrt{(1-\xi^2+\eta^2)^2+4\xi^2}}$$

6-14.exe



特点：

(1) $f(\xi,\eta)$是ξ的奇函数；

(2)每条曲线在$\xi=\eta$时取得最大值，而$\xi=-\eta$出现最小值，所以正负峰值之间距离为2ξ；

(3) 一般取η=1.5 — 3。

$$B_{pp}=2\Delta f=kf_c$$

$$\because \xi=\frac{2\Delta f}{f_c}Q,\eta=kQ$$

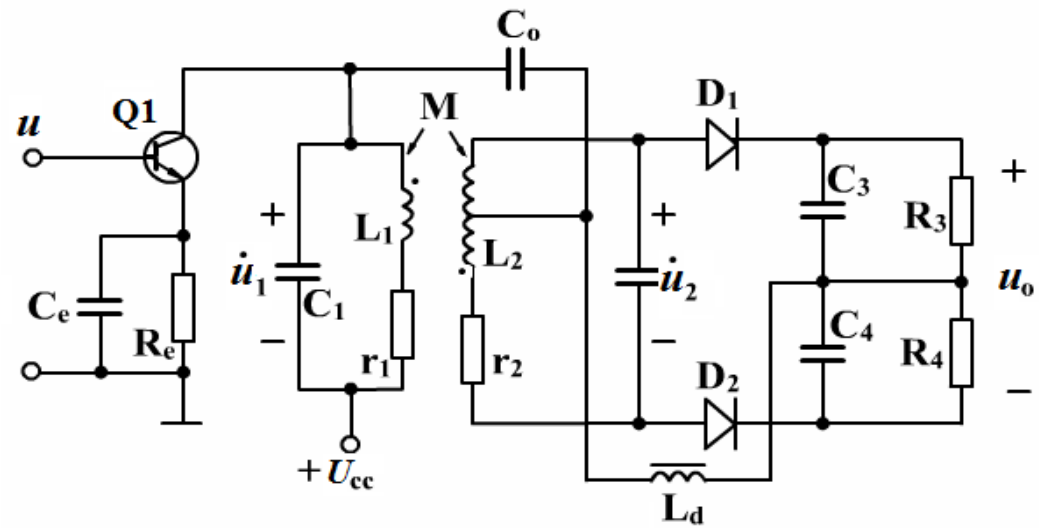
而$\xi=\eta$,

$$\therefore 2\Delta f=kf_c$$

鉴频灵敏度：$S_D=\left.\frac{du_0}{d\Delta f}\right|_{\Delta f=0}$

$$=k_dR_eI_1\frac{4\eta Q}{f_c(1+\eta^2)\sqrt{4+\eta^2}}$$

例. 相位鉴频器如图所示，若输入调频波中心频率为f_o



求:1） L_1、C_1和L_2、C_2应分别调谐于什么频率（<、>、= f_o)?

2)互感同名端如图所示，试画出鉴频特性曲线?

3）若发生下列情况，鉴频特性曲线如何变化，定性画出?

(a)D_1开路　　(b)D_2接反　　(c)D_1、D_2都接反

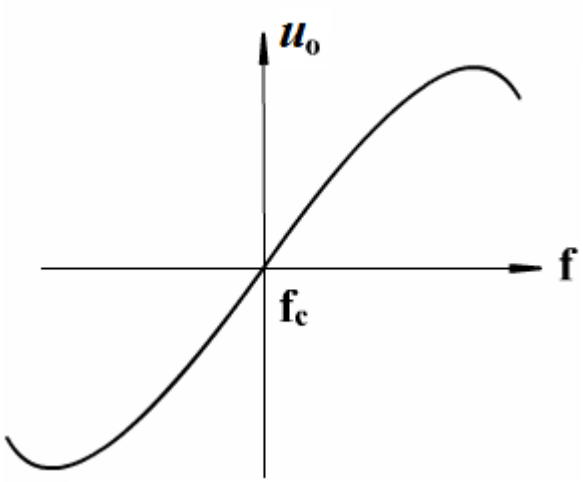
4）图中互感耦合回路的主要作用是_____

a)线性频谱搬移　b)频--相变换　c）频--幅变换　d)FM变换为AM波

解

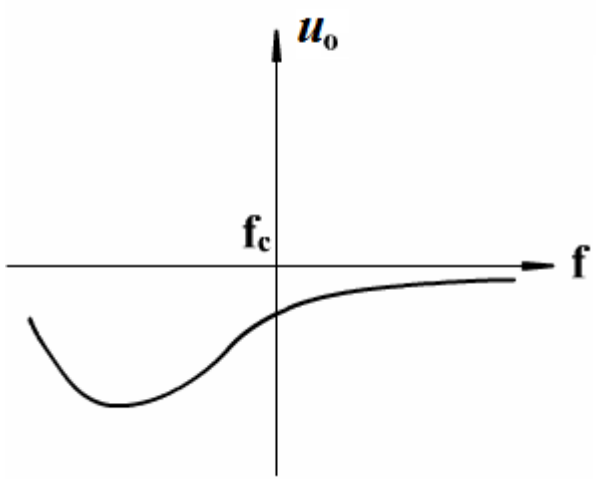
1、L_1、C_1和L_2、C_2均调谐于f_0

2、



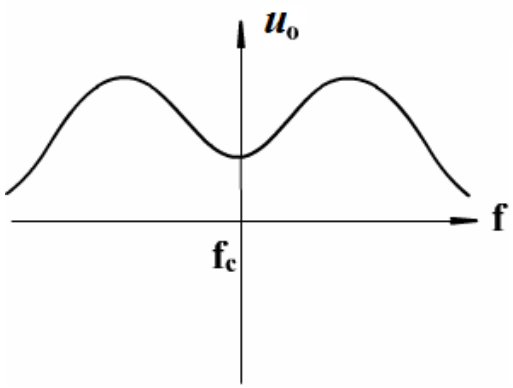
3、 (a) D_1 开路， $u_{01}=0$

$$u_0 = -k_d |u_{D2}|$$



(b) D_2反接（和比例鉴频器U_{AB}相比较）

$$u_0 = k_d(|u_{D2}| + |u_{D1}|)$$

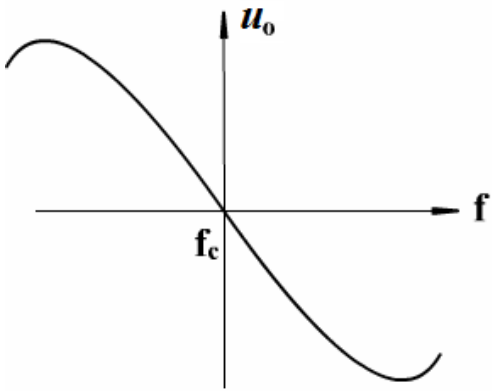


(c)　D_1、D_2都反接

$$\dot{u}_{D1} = -\dot{u}_1 - \dot{u}_2/2$$

$$\dot{u}_{D2} = -\dot{u}_1 + \dot{u}_2/2$$

$$u_0 = k_d(|u_{D2}| - |u_{D1}|)$$

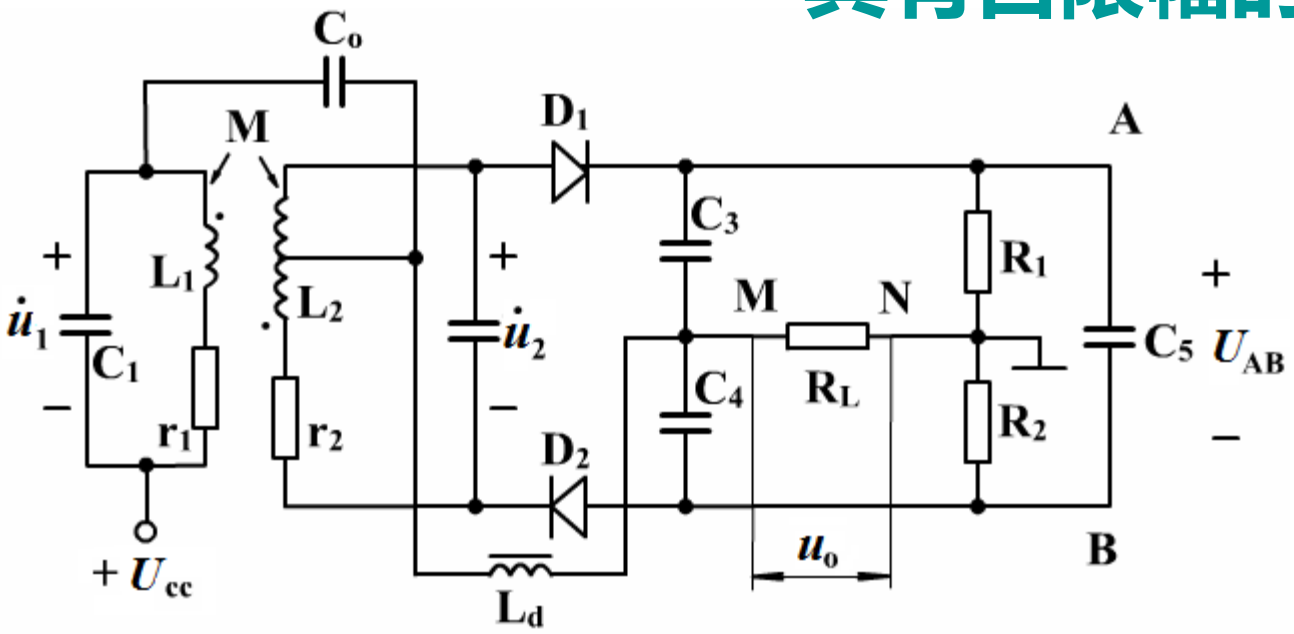


(4) 选（b）频—相变换

四. 比例鉴频器

1. 电路

具有自限幅的功能

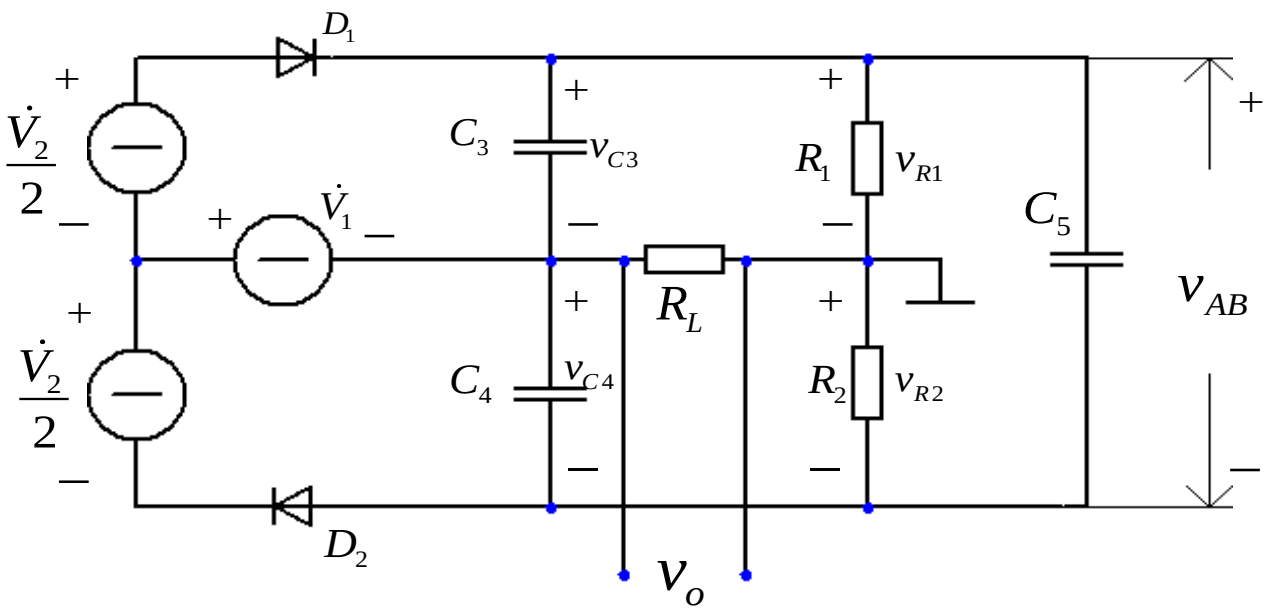


与相位鉴频器的区别：

(1) D_1 ，D_2顺接

(2) 多了大电容C_5(具有限幅的作用)

(3) 输出不同



等效电路图

对于高频来说，C_3、C_4可近似短路，故加在D_1、D_2上的高频电压为：

$$\dot{V}_{D1}=\dot{V}_1+\frac{\dot{V}_2}{2},\dot{V}_{D2}=-\dot{V}_1+\frac{\dot{V}_2}{2}$$

输出电u_o取自M、N之间，因此

$$u_0 = u_{C4} - u_{R2}$$

$$\text{或} u_0 = u_{R1} - u_{C3}$$

一般$R_1 = R_2$，则$u_{R1} = u_{R2}$，　将上两式相加，得：

$$u_0 = \frac{1}{2}(u_{C4} - u_{C3})$$

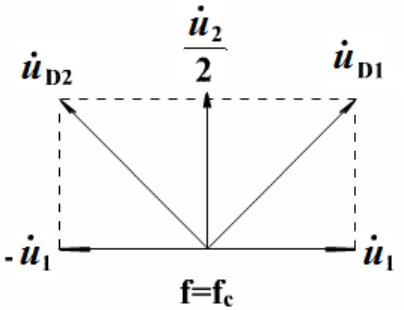
$$= \frac{1}{2}k_d(|u_{D2}| - |u_{D1}|)$$

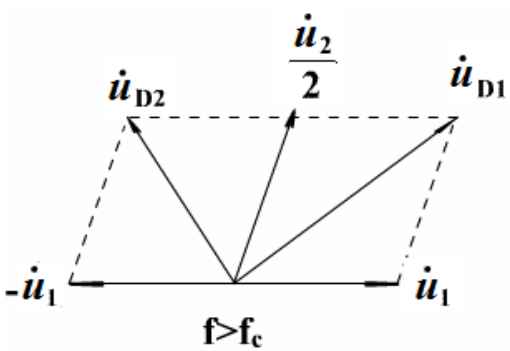
与相位鉴频器相比较：

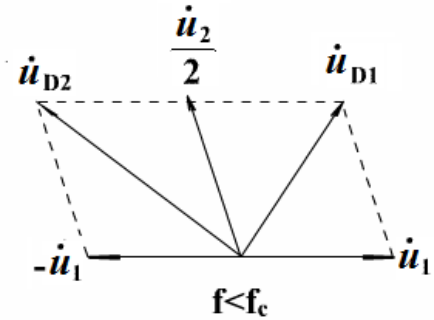
1. **输出电压极性相反；**
2. **鉴频灵敏度下降一半。**

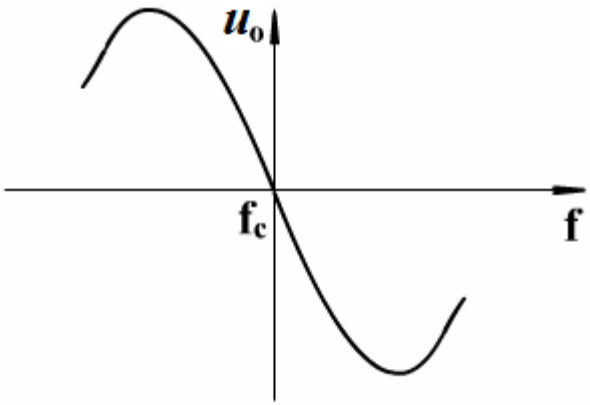
2. 鉴频特性

$$\dot{u}_{D1} = \dot{u}_1 + \dot{u}_2/2 \,, \dot{u}_{D2} = -\dot{u}_1 + \dot{u}_2/2$$









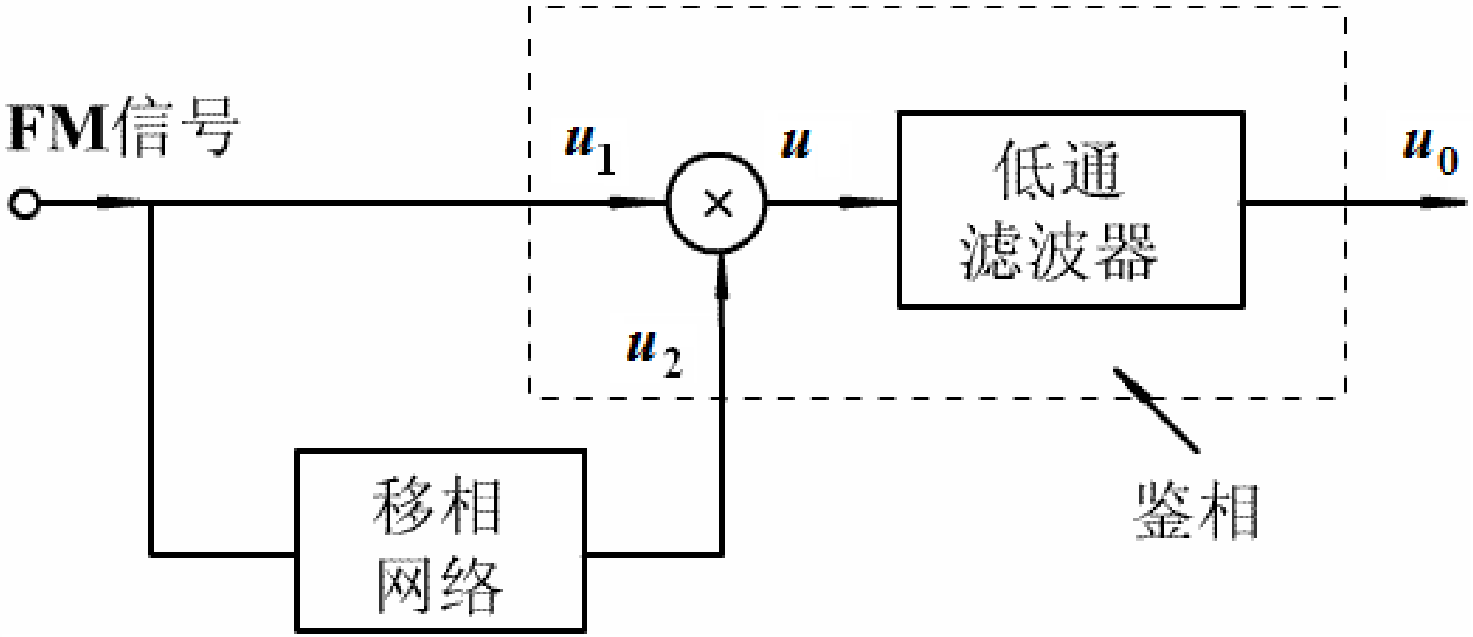
$$u_0 = \frac{1}{2} k_d \left(|u_{D2}| - |u_{D1}| \right)$$

与相位鉴频器相比较：

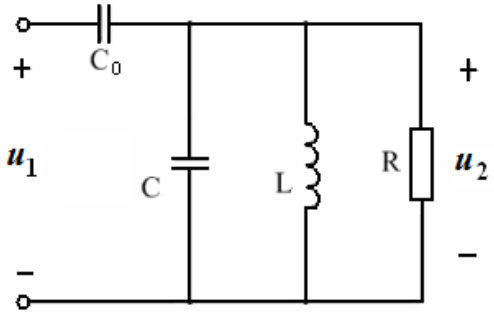
1. **输出电压极性相反；**
2. **鉴频灵敏度下降一半。**

五. 移相正交鉴频器

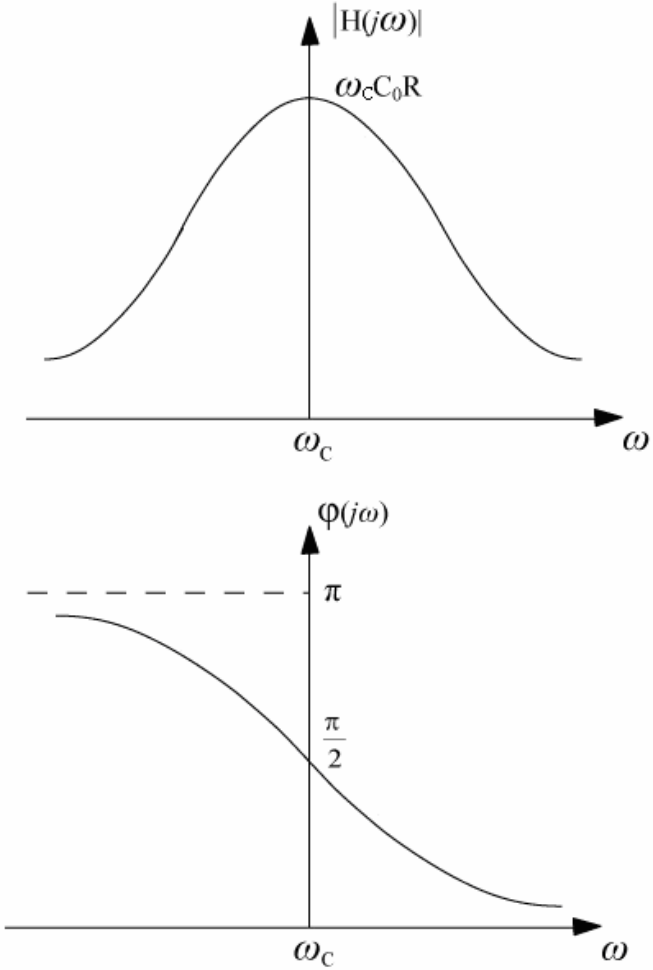
1. 原理



常用的移相电路如图所示，



(a)



(b)

$$H(j\omega)=\frac{u_2}{u_1}=\frac{\dfrac{1}{\dfrac{1}{R}+j\left(\omega C-\dfrac{1}{\omega L}\right)}}{\dfrac{1}{j\omega C_0}+\dfrac{1}{\dfrac{1}{R}+j\left(\omega C-\dfrac{1}{\omega L}\right)}}$$

$$=\frac{j\omega C_0}{\dfrac{1}{R}+j\left(\omega C_0+\omega C-\dfrac{1}{\omega L}\right)}$$

$$=\frac{j\omega C_0 R}{1+j\xi} \qquad \xi=Q\left(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega}\right)$$

$$\left|H(j\omega)\right| = \frac{\omega C_0 R}{\sqrt{1+\xi^2}} \qquad \varphi(j\omega) = \frac{\pi}{2} - \mathrm{arctg}\,\xi$$

设输入调频波 $u_1 = U_{1m}\cos\left(\omega_c t + \frac{\Delta\omega_m}{\Omega}\sin\Omega t\right)$

那么经移相网络后输出电压

$$u_2 = U_{2m}\cos\left(\omega_c t + \frac{\Delta\omega_m}{\Omega}\sin\Omega t + \varphi(j\omega)\right)$$

其中 $U_{2m} = U_{1m}\left|H(j\omega)\right|$

这两个信号加到乘法器上，设乘法器增益为K_M，乘法器输出

$$u(t)=K_M u_1 u_2$$

$$=K_M U_{1m} U_{2m}\cos\left(\omega_c t+\frac{\Delta\omega_m}{\Omega}\sin\Omega t\right)\cos\left(\omega_c t+\frac{\Delta\omega_m}{\Omega}\sin\Omega t+\varphi(j\omega)\right)$$

$$=\frac{1}{2}K_M U_{1m} U_{2m}\left[\cos(\varphi(j\omega))+\cos\left(2\omega_c t+2\frac{\Delta\omega_m}{\Omega}\sin\Omega t+\varphi(j\omega)\right)\right]$$

经低通滤波器后，设滤波器增益为K_F

$$u_0(t)=\frac{1}{2}K_M K_F U_{1m} U_{2m}\cos(\varphi(j\omega))$$

$$=\frac{1}{2}K_M K_F U_{1m} U_{2m}\cos\left(\frac{\pi}{2}-\mathrm{arctg}\,\xi\right)$$

令 $\theta=\mathrm{arctg}\,\xi$

$$u_0(t)=\frac{1}{2}K_F K_M U_{1m} U_{2m}\cos\left(\frac{\pi}{2}-\theta\right)$$

$$=\frac{1}{2}K_F K_M U_{1m} U_{2m}\sin\theta$$

当$\theta \leqslant \frac{\pi}{6}$时， $\sin\theta \approx \theta = \mathrm{arctg}\,\xi \approx \xi \approx 2Q_L \dfrac{\Delta\omega}{\omega_c}$

这时，有

$$u_0(t) = \frac{1}{2} K_F K_M U_{1m} U_{2m} \cdot 2Q_L \frac{\Delta\omega}{\omega_c}$$

$$= K_F K_M U_{1m} U_{2m} Q_L \frac{\Delta\omega}{\omega_c}$$

鉴频器的输出与输入信号的频偏成正比

数字调频与调相

二进制频移键控(FSK)调制与解调

二进制移相键控(PSK)调制与解调